the following chapter gives

$$\mathcal{R}_0 = \frac{c}{\lambda}\left(\frac{c}{2l}\Big/\frac{\pi\sqrt{R}}{1-R}\right)^{-1}$$
$$= 2\pi l\sqrt{R}/\lambda(1-R). \qquad (2.50)$$

This is the same as equation (2.49) within a factor $\sqrt{R}$, which is accounted for by the small-loss approximation used in deriving Q. Thus, as expected, the optical resolving power is synonymous with quality factor Q.

This comparison further establishes that if the energy losses are due entirely to reflection-like losses at the mirrors (i.e. transmission, scattering and absorption) the resolving power may be rigorously derived according to the ray treatment. If however losses due to diffraction become appreciable the effective resolving power must decrease as Q itself decreases. This will become important only for small aperture and Fresnel numbers of $\leqslant 10^2$. In typical spectroscopic operation as we have seen the Fresnel number is of order 10^5.

The broad conclusion is that, while cavity mode analysis is essential for low Fresnel numbers and for coherent light applications involving mode matching etc, for most spectroscopic purposes involving large Fresnel number systems the geometric ray treatment is perfectly adequate. In particular this permits the straightforward evaluation of instrumental form, light gathering and practical operation and is developed at length in the following chapter for the plane parallel-plate interferometer.

Various aspects of these matters have been interestingly discussed in several papers including Barone (1963), Fox and Li (1961), Boyd and Gordon (1961) and the reviews already listed.

From: J.M. Vaughan "The Fabry-Perot Interferometer" (1989) Chapter 3

3 The Plane Fabry–Perot Interferometer

3.1 Introduction

In this chapter a detailed development is given of the theory and practice of the plane interferometer on the basis of the simple ray treatment. Many descriptions exist in the literature from the earliest papers of Fabry and Perot (1897, 1899), valuable review articles of e.g. Meissner (1941), Kuhn (1951) and Jacquinot (1960) and texts such as Tolansky (1947), Steel (1967) and Born and Wolf (1975). In this present chapter a very deliberate attempt has been made to key the discussion to example and illustration so that it serves as a practical guide. The important parameters of light gathering, finesse, resolving power, overlap of orders, scanning aperture, etc are described and also considerations due to the different types of light source. The use of the interferometer with photoelectric and photographic recording is outlined, and also some aspects of the analysis of spectra, the effect of plate imperfections and deconvolution. However detailed discussion of the design, construction and operation of plane interferometers is deferred to the following chapter.

3.2 General form of the interference fringes

3.2.1 Geometric ray treatment

In ordinary spectroscopic use the significant questions one asks of an interferometer are the dependence on angle and wavelength of the light transmitted through the instrument. The distribution of field over the plates, requiring cavity mode analysis, is not usually considered and as discussed in the previous chapter has received sophisticated treatment only with the advent of the laser. Thus for spectroscopic purposes the geometric treatment is usually sufficient and probably more instructive (e.g. Kuhn 1951, Steel 1967, Born and Wolf 1975).

Consider the plane waves shown successively reflected between plane parallel

surfaces ① and ② in figure 3.1(*a*). The incident ray i of unit amplitude represents the direction of propagation of the incident plane wave and θ is the angle of incidence of the beams on the surfaces from within the interspace, as shown. The light is supposed to be of wavelength λ_0 *in vacuo*, and the effective refractive index of the interspace is μ. For convenience the electric vector of the incident wave is taken to be of unit amplitude, and is assumed to be linearly polarized either parallel or perpendicular to the plane of incidence.

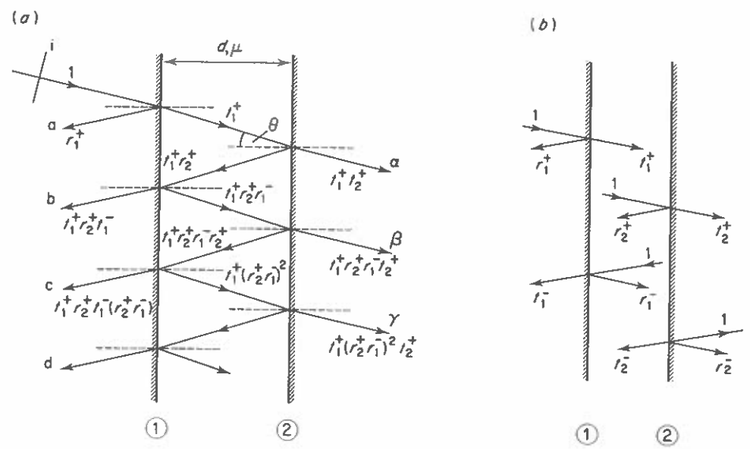

Figure 3.1 (*a*) Plane waves successively reflected back and fore between two plane parallel surfaces. The relative amplitude coefficients of the reflected set of beams a, b, c, d, . . . and transmitted beams α, β, γ, . . . are shown. (*b*) An illustration of the notation used for the complex reflection and transmission amplitude coefficients. The reflected and transmitted beams arising from an incident beam travelling left to right are labelled positive while those due to an incident beam travelling right to left are labelled negative.

Unfortunately there seems to be no accepted convention in labelling the amplitude coefficients of transmission and reflection at the various surfaces. Furthermore the nomenclature found in a number of Airy formula derivations (e.g. Kuhn 1951, Born and Wolf 1975, Steel 1967) is orthogonal to that often used in the discussion of multilayer thin films (e.g. Liddell 1981). For the present the convention shown in figure 3.1(*b*) will be used in which the surface reflection and transmission coefficients for an incident ray travelling left to right are labelled + (positive-going incident) and for an incident ray travelling right to left are labelled − (negative-going incident). This convention in fact follows Macleod (1986) and has the virtue of easy extension to three or more surfaces. The coefficients are supposed complex and for the moment we make no assumptions as to their magnitude.

We now consider the sets of successively reflected waves a, b, c, . . . and

transmitted waves α, β, γ, . . . shown in figure 3.1(*a*). With the convention described above their respective amplitudes are labelled as shown. The optical delay for successive reflections gives an additional phase difference which for adjacent members of each set obviously corresponds to a double passage of the interspace. This phase lag φ is easily shown to be given by

$$\varphi = 2\pi(2\mu d\cos\theta)/\lambda_0. \tag{3.1}$$

Thus with the exception of the first reflection a, both sets of beams form geometrical progressions with the same factor $r_1^- r_2^+ e^{i\varphi}$. If the first m transmitted beams are now superposed, the amplitude $E_t(m)$ of the resultant electric vector of transmitted light is given by

$$\begin{aligned} E_t(m) &= t_1^+ t_2^+[1 + r_1^- r_2^+ e^{i\varphi} + \ldots + (r_1^- r_2^+)^{m-1} e^{i(m-1)\varphi}] \\ &= t_1^+ t_2^+[1 - (r_1^- r_2^+)^m e^{im\varphi}]/(1 - r_1^- r_2^+ e^{i\varphi}). \end{aligned} \tag{3.2}$$

For an infinite number of beams as $m \to \infty$ this becomes

$$E_t \to E_t(\infty) = t_1^+ t_2^+/(1 - r_1^- r_2^+ e^{i\varphi}). \tag{3.3}$$

The corresponding transmitted intensity I_t is found from the square modulus $E_t E_t^*$ and

$$I_t = |t_1^+ t_2^+|^2/(1 + |r_1^- r_2^+|^2 - 2|r_1^- r_2^+|\cos\psi). \tag{3.4}$$

In this expression

$$\psi = \varphi + \varepsilon \tag{3.5}$$

where ε is the correction for phase changes on reflection given by

$$\varepsilon = \arg r_1^- + \arg r_2^+. \tag{3.6}$$

The expression (3.4) needs to be interpreted with some care. If the surfaces ① and ② are identical uncoated dielectrics, the coefficients r and t are real and given in terms of the angles and refractive indices by the usual Fresnel formulae (see e.g. Born and Wolf (1975, §1.5.2) or Macleod (1986, chapter 2). For a single surface these may be summarized by

$$t^+ t^- = T \qquad r^+ = -r^- \qquad r^{+2} = r^{-2} = R \qquad \text{and } R + T = 1 \tag{3.7}$$

where R and T are the surface intensity reflection and transmission coefficients. On substituting in (3.4), since $\varepsilon = 0$ and recognizing that for identical dielectrics $t_2^+ = t_1^-$ and $r_2^+ = r_1^-$, the energy transmission coefficient for the pair of surfaces becomes

$$\begin{aligned} I_T &= T^2/(1 + R^2 - 2R\cos\varphi) && (3.8a) \\ &= T^2/[(1 - R)^2 + 4R\sin^2(\varphi/2)] && (3.8b) \\ &= \frac{T^2}{(1 - R)^2}\left(\frac{1}{1 + [4R/(1 - R)^2]\sin^2(\varphi/2)}\right) && (3.8c) \\ &= [T/(1 - R)]^2[1 + F\sin^2(\varphi/2)]^{-1} && (3.8d) \\ &= [T/(1 - R)]^2 \mathcal{A}(\varphi) && (3.8e) \end{aligned}$$

where F is given by $4R/(1-R)^2$. Equations (3.8) are commonly used expressions of the basic Airy formula for transmission, and $\mathcal{A}(\varphi)$ is called the Airy shape function.

In considering the reflected light, if the first m reflected beams are superimposed, the amplitude $E_r(m)$ of the resultant electric vector is given by

$$E_r(m) = r_1^+ + t_1^+t_1^-r_2^+e^{i\varphi}[1 + r_1^-r_2^+e^{i\varphi} + \ldots + (r_1^-r_2^+)^{m-2}e^{i(m-2)\varphi}] \quad (3.9)$$

$$= r_1^+ + \{t_1^+t_1^-r_2^+e^{i\varphi}\,[1 - r_1^-r_2^+e^{i(m-1)\varphi}]\}/(1 - r_1^-r_2^+e^{i\varphi}). \quad (3.10)$$

For an infinite number of beams as $m \to \infty$ this becomes

$$E_r \to E_r(\infty) = r_1^+ + (t_1^+t_1^-r_2^+e^{i\varphi})/(1 - r_1^-r_2^+e^{i\varphi}). \quad (3.11)$$

The reflected intensity coefficient may be evaluated as before. However in general the properties of the reflected beam are quite complicated; this comes about because the phase difference between a, the first reflected beam, and the following beam b is quite different from that between succeeding pairs b–c, c–d etc. Inspection of (3.3) and (3.11) shows that in fact the summation of waves b, c, d, ... is of the same form as the transmitted beams α, β, γ, ... In consequence, for values of $\psi = 2\pi n$ (equation (3.5)) for which the transmitted light has maximum intensity, the sum of b, c, d, ... also has a maximum; the beams are all in phase. Some general properties of the reflected fringe system are discussed in the following section. For the moment it is worth pursuing the particular case of two identical dielectric surfaces. Using the relations in equation (3.7) and forming $E_rE_r^*$ gives the intensity of the reflected beam as before. In this case equation (3.11) may be rewritten as

$$\begin{aligned} E_r &= r_1^+ - t_1^+t_1^-r_1^+e^{i\varphi}/(1 - r_1^-e^{i\varphi}) \\ &= \sqrt{R}(1 - Re^{i\varphi} - Te^{i\varphi})/(1 - Re^{i\varphi}) \\ &= \sqrt{R}(1 - e^{i\varphi})/(1 - Re^{i\varphi}) \end{aligned} \quad (3.12)$$

using equation (3.7). Forming the complex conjugate gives for the reflected light

$$\begin{aligned} I_R = E_rE_r^* &= R(2 - 2\cos\varphi)/(1 + R^2 - 2R\cos\varphi) \\ &= 4R\sin^2(\varphi/2)/[(1-R)^2 + 4R\sin^2(\varphi/2)] \quad (3.13a) \\ &= F\sin^2(\varphi/2)[1 + F\sin^2(\varphi/2)]^{-1}. \quad (3.13b) \end{aligned}$$

Examination of these expressions for the transmitted and reflected patterns, in particular (3.8d) and (3.13b), shows that they are complementary in that

$$I_T + I_R = [T^2/(1-R)^2 + F\sin^2(\varphi/2)]/[1 + F\sin^2(\varphi/2)]^{-1} \quad (3.14)$$

which is equal to 1 for the case of no surface absorption with $R + T = 1$. In fact this must obviously be true since, as emphasized by Kuhn (1951), radiation incident within a given solid angle remains within the corresponding elements of solid angle in the reflected and transmitted beams. If no energy is lost at the surfaces by absorption or scattering the patterns must be complementary.

The shape of the fringe patterns may be examined by considering I_T and I_R as

a function of phase difference φ. For typical dielectrics at near-normal incidence R is small ($\leqslant 0.1$) and accordingly F is small. As a first approximation, on expanding the expressions and retaining only the first term in F, these become

$$I_R \simeq F\sin^2(\varphi/2) = F(1 - \cos\varphi)/2 \quad (3.15)$$

$$I_T \simeq 1 - F\sin^2(\varphi/2) = 1 - F(1 - \cos\varphi)/2 \quad (3.16)$$

which are characteristic of only two interfering beams. Thus for low reflectivities the fringes are broad and only in the reflected system do the intensity minima approach zero. For increasing values of R the value of F rapidly becomes larger (e.g. $R = 0.90$, $F = 360$) and the fringes become sharper. In the transmitted system there is appreciable light only in the immediate vicinity of the maxima. Thus the transmitted fringe pattern consists of sharp bright fringes against a greatly reduced background. Correspondingly the reflected pattern is one of narrow dark fringes on a bright background.

High reflectivities at near-normal incidence are commonly achieved using multiple layers of thin dielectric films or partially transparent metal films (§2.6). For metal films the reflectivities and phase changes at the surfaces are no longer simply related and there is absorption (for discussion of the theory of metal films on dielectric substrates see e.g. Born and Wolf 1975). The question is then how well do equations (3.8) and (3.13) represent the situation? In transmission, following from equations (3.4) and (3.5), and providing that the two surfaces of the plate are identical (so that $r_2^+ = r_1^-$), the intensity transmission coefficient is given by

$$\begin{aligned} I_T &= [T/(1-R)]^2/[1 + F\sin^2(\psi/2)] \quad (3.17) \\ &= [(1 - R - A)/(1-R)]^2/[1 + F\sin^2(\psi/2)] \quad (3.18) \\ &= [1 - A/(1-R)]^2\mathcal{A}(\psi) \quad (3.18a) \end{aligned}$$

where $R = |r_2^+|^2 = |r_1^-|^2$ is the surface reflectivity from within the interspace, ψ is equal to $\varphi + \varepsilon$ and thus includes the surface phase changes (ε is equal to $2\arg r_2^+$), and A, the fraction of light absorbed by the metal, is given by

$$R + T + A = 1. \quad (3.19)$$

For dielectric films the form of the fringe is similarly given by (3.18), but the term A is largely replaced by A_s due to losses by scattering. Thus the principle effect on the transmitted fringes for surfaces having absorption and/or scatter is to reduce the fringe intensity by the factor $[1 - A/(1-R)]^2$ or equivalently $[1 - A/(A+T)]^2$; obviously light losses can only be kept small if A is small compared with T. The transmitted fringes are discussed at greater length in §3.2.3. In summary, referring to equation (3.18), variation of φ is due to different terms: for colour filters the variation is produced by the factor $1/\lambda_0$; for Fabry–Perot fringes with monochromatic light by variation of θ in photographic use or μ or d in photometric scanning; and for wedge-like Fizeau fringes by relative change of spacing across the area of the plates or surfaces. Fizeau fringes in transmission and reflection are outlined briefly in Appendix 3.

The reflected fringe system with metal films shows considerable complexity and as indicated depends critically on the various phase changes at the input surface. This is discussed in the following section.

3.2.2 The reflected fringe system

Reflected interference fringes have been widely used in metrology and investigations of surface structure (e.g. Tolansky 1948). The first investigation of such fringes appears to have been carried out by Hamy in 1906 using chemically deposited silver films. A detailed study, confirming and extending Hamy's observations, was reported by Holden in 1949 employing evaporated silver films. The present account is largely based on Holden's paper.

As indicated the form of the reflected fringe system and its relation to the transmitted system depends critically on the surface phase changes. For clarity it is probably worth labelling these individually. As a matter of convenience we employ the nomenclature used by Holden and shown in figure 3.2. Thus in relation to figure 3.1 and equation (3.11)

$$r_1^+ = r_1^+ e^{i\alpha} \qquad r_2^+ = r_2^+ e^{i\beta_2} \qquad r_1^- = r_1^- e^{i\beta_1} \qquad \text{etc.} \tag{3.20}$$

In equation (3.9) the mth reflected beam is given by

$$t_1^+ t_1^- r_2^+ \, e^{i(\varphi+2\tau_1+\beta_2)}[(r_1^- r_2^+)^{m-2} e^{i(m-2)(\varphi+\beta_1+\beta_2)}]$$

and substituting in equation (3.11) one obtains for the reflected amplitude coefficient

$$E_r = r_1^+ e^{i\alpha} + (t_1^+ t_1^- r_2^+ \, e^{i(\varphi+2\tau_1+\beta_2)})/(1 - r_1^- r_2^+ \, e^{i(\varphi+\beta_1+\beta_2)}) \tag{3.21}$$

$$= r_1^+ e^{i\alpha} + \frac{t_1^+ t_1^- r_2^+ (e^{i(\varphi+2\tau_1+\beta_2)} - r_1^- r_2^+ e^{i(2\tau_1-\beta_1)})}{1 + (r_1^- r_2^+)^2 - 2r_1^- r_2^+ \cos(\varphi+\beta_1-\beta_2)}. \tag{3.22}$$

By taking the product $E_r E_r^*$, with careful accounting of cross terms we obtain the reflected intensity

$$I_R = (r_1^+)^2 + \frac{(t_1^+ t_1^- r_2^+)^2[1 + (r_1^- r_2^+)^2 - 2r_1^- r_2^+ \cos(\varphi + \beta_1 + \beta_2)]}{[1 + (r_1^- r_2^+)^2 - 2r_1^- r_2^+ \cos(\varphi + \beta_1 + \beta_2)]^2}$$

$$+ \frac{2t_1^+ t_1^- r_2^+ r_1^+ [\cos(\varphi + 2\tau_1 + \beta_2 - \alpha) - r_1^- r_2^+ \cos(2\tau_1 - \beta_1 - \alpha)]}{[1 + (r_1^- r_2^+)^2 - 2r_1^- r_2^+ \cos(\varphi + \beta_1 + \beta_2)]}. \tag{3.23}$$

From equation (3.4) the transmitted intensity is given by

$$I_T = (t_1^+ t_2^+)^2/[1 + (r_1^- r_2^+)^2 - 2r_1^- r_2^+ \cos\psi]$$

$$= [(t_1^+ t_2^+)^2/(1 - r_1^- r_2^+)^2] \mathcal{A}(\tau) \tag{3.24}$$

where ψ is the optical separation of the plates, equal to $\varphi + \beta_1 + \beta_2$, and $\mathcal{A}(\psi)$ is the Airy shape function. In equation (3.23) we set $(2\tau_1 - \beta_1 - \alpha)$ equal to χ and note that this term depends only on the phase conditions at the input surface 1. Substituting in (3.23) we have for the reflected system

$$I_R = (r_1^+)^2 + [\mathcal{A}(\psi)/(1 - r_1^- r_2^+)^2]$$
$$\times [(t_1^+ t_1^- r_2^+)^2 + 2t_1^+ t_1^- r_2^+ r_1^+ \cos(\psi + \chi) - 2(t_1^+ t_1^- r_2^+ r_1^+)(r_1^- r_2^+) \cos\chi]. \tag{3.25}$$

Comparison of this expression with equation (3.24) shows that the reflected system, in addition to the background term $(r_1^+)^2$, has two extra terms that modify the 'transmission-like' Airy function. One of these is of constant value $\cos\chi$, but the other, of form $\cos(\psi + \chi)$, is obviously out of phase with ψ by χ.

Holden has analyzed this situation in some detail; in general the reflected fringes are not symmetrical. Indeed measurement of the asymmetry, against the symmetrical transmitted system for the same films, gives an experimental determination of χ for the input film. It is readily shown that there are two principal limiting cases, with $\chi = 2n\pi$ and $(2n + 1)\pi$, when symmetrical fringes may be expected.

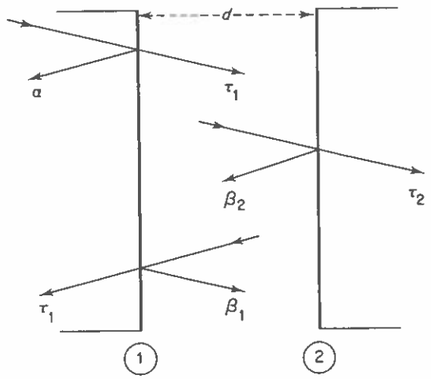

Figure 3.2 A diagram illustrating the phase changes on reflection and transmission, employing for convenience the notation of Holden (1949). The phase changes are relative to the physical surfaces supposed separated by a distance d.

Experimental observations were made by Holden with carefully prepared silver films of varying reflectivity up to 90%. For best comparison of the transmitted and reflected systems, Fizeau-type wedge fringes were employed. As outlined in Appendix 3 phase lag terms due to the wedge angle can create asymmetry and subsidiary maxima. However, by appropriate choice of wedge angle and order of interference, these can be made negligibly small in comparison with those due to the intrinsic phase characteristics of the first surface. The experimental observations may be summarized as follows.

(i) With the first surface uncoated, so that $r_1^2 = 0.04$ and $\chi = (2n + 1)\pi$, the reflected minima are coincident with transmitted maxima.

(ii) As the first surface reflectivity is increased the fringes go through an asymmetrical sequence until at about 15% the reflected fringes show quite sharp maxima almost coincident with the transmission maxima, but much more luminous. This is illustrated in figure 3.3; such fringes were described by Hamy. For silver films and a near-opaque second surface, Holden reported that these reflected fringes had a half-width as small as one-fifth of the distance between orders for green light. From the symmetry it is apparent that χ must be close to $2n\pi$ in this region.

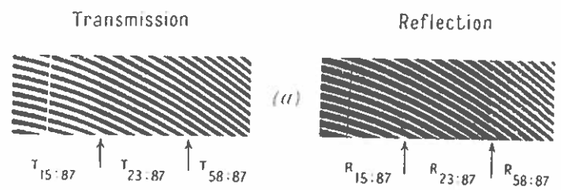

Figure 3.3 Transmission and reflection fringes in a narrow wedge between two silvered surfaces. The reflectivities of different areas are indicated. Note that for reflectivities 15 : 87 the reflection maxima are symmetric and sharper than the transmission maxima with which they coincide. In $R_{58:87}$ the asymmetry of the reflected fringes is clearly evident for higher reflectivities of the first surface. (From Holden 1949, figure 5(*a*).) Reproduced by permission of IOP Publishing Ltd.

(iii) As the reflectivity is increased further above 15% the fringes develop an asymmetric maximum–minimum form until at about 60% the minimum is nearly symmetric and increases in sharpness up to the highest reflectivities of 90%. This implies that χ is tending to $(2n + 1)\pi$.

Reflectivities greater than 60% are commonly used in topographical studies and it is of interest to examine the form of the fringes. We consider a symmetrical arrangement with identical high-reflection coatings; writing $r^2 = R$, $t^2 = T$ and taking $\chi = (2n + 1)\pi$, the reflected system, equation (3.25) becomes in various forms:

$$I_R = R + [\mathcal{A}(\psi)/(1 - R)^2](T^2R + 2TR^2 - 2TR\cos\psi) \tag{3.26a}$$

$$= R + [\mathcal{A}(\psi)/(1 - R)^2][T(1 + R^2 - 2R\cos\psi) - T(1 - R^2 - RT)] \tag{3.26b}$$

$$= (R + T) - [\mathcal{A}(\psi)/(1 - R)^2]T[1 - R(R + T)] \tag{3.26c}$$

$$= (1 - A) - (1 - A)\mathcal{A}(\psi) + A^2R\mathcal{A}(\psi)/(1 - R)^2. \tag{3.26d}$$

Maxima thus occur at $\psi = (2n + 1)\pi$ and the minima at $\psi = 2n\pi$. From (3.26*c*) the minimum intensity is given by

$$[I_R]_{\min} = (R + T) - T[1 - R(R + T)]/(1 - R)^2. \tag{3.27}$$

If there is no absorption in the films, setting $R + T = 1$ in equation (3.27) shows that the minima are of zero intensity. The maxima of intensity may

correspondingly be shown to be $[1 - (1 - R)^2/(1 + R)^2]$, and as expected (equation (3.14)) the reflected fringe system is complementary to the transmitted one with

$$I_R = 1 - [\mathcal{A}(\psi)/(1 - R)^2]T^2 = 1 - I_T. \tag{3.28}$$

In the presence of absorption A, defined by $R + T + A = 1$, the minimum reflected intensity is no longer zero but rises to $A^2R/(1 - R)^2$, equal to $A^2R/(A + T)^2$, and the maxima are reduced by $\sim 4AR/(1 + R)^2$.

For purposes of metrology and topographical studies Holden evaluated the reflected fringe characteristics. Most importantly he found that the reflected dark fringes are notably sharper than transmitted fringes for R greater than 80%. However at the upper end of the range, $R \sim 90\%$, their visibility is critically dependent on the value of A. This is immediately obvious from the above expressions since the intensity in the fringe minimum increases sharply when A becomes comparable to T. In addition for high reflectivities the collimation conditions on the illumination become quite severe.

Further information on reflected fringe systems between parallel or near-parallel surfaces may be found in Holden's paper or the numerous books and publications of Tolansky. Reflected fringes are not discussed further in this volume, other than a brief reference in Chapter 6 on multiple interferometers and Appendix 3 on Fizeau fringes.

3.2.3 The transmitted fringe system

Equations (3.4), (3.8) and (3.18) show various expressions of the basic Airy formulae for the transmitted light. These have been derived by superposition and coherent addition of beams successively reflected between the two plane parallel surfaces. In practice such superposition is usually achieved in the focal plane of a 'fringe forming' lens. Figure 3.4 shows a basic arrangement in which light from a source is collected by a lens L_1 and passed into the interferometer. Lens L_2 gathers the transmitted beams and brings them together at the focal plane as shown. The two principal optical surfaces are worked in high-quality optical materials of sufficient thickness relative to their size to ensure good long-term stability. Plates individually flat to better than $\lambda/150$ in the visible region over 5 cm working diameter are commonly available; plates of over 12 cm working diameter have been employed. The reflecting surfaces are coated with partially transmitting films of silver, aluminium, gold or more commonly since ~1960, dielectric materials (§2.6).

With the arrangement shown in figure 3.4, with a light source of quasi-monochromatic wavelength λ_0 we suppose that, in the absence of the interferometer, the illumination at the focal plane of L_2 is given by I_0. When the interferometer is inserted the illumination or intensity distribution (more strictly the brightness) is determined by the Airy transmission coefficient and given by

$$I(\psi) = I_T I_0 = I_0[T/(1 - R)]^2/[1 + F\sin^2(\psi/2)] \tag{3.29}$$

where the coefficient F is given by

$$F = 4R/(1 - R)^2 \quad \text{and } \psi = \varphi + \varepsilon. \tag{3.30}$$

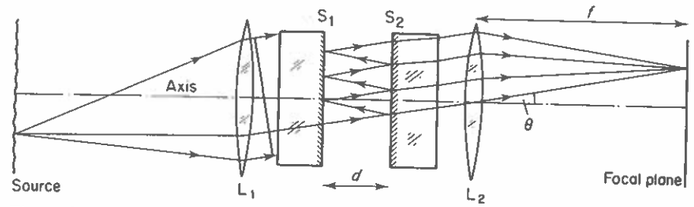

Figure 3.4 An illustration of the basic interferometric arrangement with plane parallel optical surfaces S_1 and S_2. Light from the source is collected by lens L_1. Successively reflected waves are superposed by lens L_2 and fringes are formed in the focal plane.

Maxima in the transmitted light thus occur for values of $\psi = 2\pi n$ where n is an integer. From the cylindrical symmetry of the arrangement about the optic axis, the fringe pattern obviously consists of circular bright rings, as first observed by Haidinger (Chapter 1), of angular radius θ given by

$$\psi = 2\pi n = 2\pi(2\mu d \cos\theta)/\lambda_0 + \varepsilon \tag{3.31}$$

whence

$$2\mu d \cos\theta = \lambda_0(n - \varepsilon/2\pi). \tag{3.32}$$

From these equations the phase changes on reflection have no effect on the shape of the intensity distribution but serve merely to displace the fringes. This is equivalent to a small change in the effective value of the étalon spacing d amounting to $(\lambda_0/2\mu\cos\theta)(\varepsilon/2\pi) \leqslant \lambda/2$. For higher orders of interference, with $n \geqslant 100$, this correction can be ignored for most practical purposes. This is certainly true for many spectroscopic applications where the exact order of interference is not required (for example measurement of line profile, or the frequency separation of a number of close spectral components). It may not be true for precise inter-comparison of standard wavelengths separated sufficiently in wavelength that the phase changes on reflection have altered. However as a matter of convenience the phase changes occurring on reflection will for the remainder of this discussion be treated as contributing to the effective optical separation of the plates d'. Consequently in equation (3.29) ψ is replaced by φ and equation (3.31) for the bright rings becomes

$$\varphi_n = 2\pi n = 2\pi(2\mu d' \cos\theta_n)/\lambda_0 \tag{3.33}$$

and

$$\theta_n = \cos^{-1}(\lambda_0/2\mu d')n. \tag{3.34}$$

In many instances the distinction between d and d' will be ignored. From the arrangement shown in figure 3.4 the circular fringes are obviously of radius $f\theta_n$, where f is the focal length of L_2, and they are centred on the optic axis. The fringes become sharper for higher values of the reflectivity R and their half-width $\pm\varphi_{1/2}$ at half-intensity is given by

$$F\sin^2(\varphi_{1/2}/2) = 1. \tag{3.35}$$

For F sufficiently large the sine may be approximated by the angle and

$$(\varphi_{1/2}/2)^2 = F^{-1} \quad \text{and } \varphi_{1/2} = (1 - R)/\sqrt{R}. \tag{3.36}$$

The change of phase φ between successive orders is 2π, thus the reciprocal of the full width of a fringe at half-intensity expressed as a fraction of an order is given by

$$\begin{aligned} N_R &= 2\pi/2\varphi_{1/2} = \pi\sqrt{R}/(1 - R) \\ &= (\pi/2)\sqrt{F}. \end{aligned} \tag{3.37}$$

This term N_R is called the *ideal reflectivity finesse* of the interferometer and is discussed at length in the following sections.

It is worth appreciating the high efficiency of the interferometer considered as an angular filter. From equation (3.29) the transmitted intensity at the fringe peak is seen to be $I_0T^2/(1 - R)^2$. For perfect non-absorbing and non-scattering films, $T + R = 1$ so that the illumination at positions corresponding to the peak of the fringes is unchanged by insertion of the interferometer. There is of course no element of amplification in this, or of transferring flux from one part of the image to another; the plane interferometer serves only as an angle-dependent filter. As shown in §3.2.1 absorption or scatter loss A in the reflecting surfaces reduces the fringe intensity which at the peak is given by

$$I_{max} = I_0[1 - A/(A + T)]^2 = I_0T^2/(1 - R)^2 \tag{3.38}$$

and in the mid-order fringe minimum or trough by

$$\begin{aligned} I_{min} &= I_0[1 - A/(A + T)]^2/(1 + F) && (3.39) \\ &= I_0[1 - A/(A + T)]^2/[(1 + R)^2/(1 - R)^2] && (3.40) \\ &= I_0[T/(1 - R)]^2/[(1 + R)^2/(1 - R)^2] && (3.40a) \\ &= I_0T^2/(1 + R)^2. && (3.40b) \end{aligned}$$

Thus the ratio of peak to background, usually called the extinction or contrast ratio, is given by $(1 + R)^2/(1 - R)^2$. Typically for $R \simeq 0.94$ the corresponding extinction is $\sim 10^3$. While quite high, for many purposes, particularly the study of weak spectral features in the presence of bright ones, this is often not high enough. Many techniques of using several étalons in series, or of multi-passing a single étalon to improve contrast have been developed and are described in Chapter 6. A graphical compilation of finesse, transmission coefficient and extinction for different values of A, R and T is also shown in figure 6.11.

The periodicity of the fringes clearly arises from the $\sin^2(\varphi/2)$ term in the denominator of the Airy formula; the form of the fringes is shown in figures 3.5 and 3.6. In the first of these the fringe profiles are drawn for a uniformly changing abscissa φ. Regularly spaced fringes appear with maxima at $\varphi = 2\pi n$, $2\pi(n - 1)$, $2\pi(n - 2)$, ... These diagrams show the intensity that would be found at a fixed point in the fringe pattern (θ constant) as the optical path separation $2\mu d$ of the étalon plates is uniformly changed.

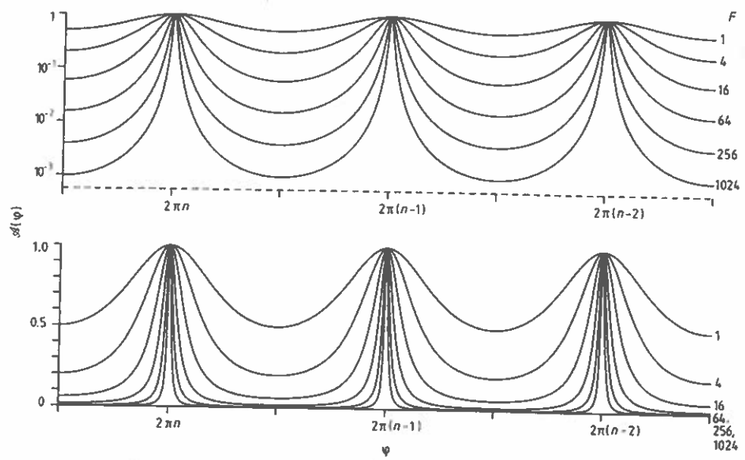

Figure 3.5 Fringe profiles of $\mathscr{A}(\varphi)$ the Airy shape function (equation (3.8*e*)) shown on both log and linear scales for a uniform change of φ and at various values of F, where the finesse $N_R = (\pi/2)\sqrt{F}$.

In contrast figure 3.6 shows the intensity profile along a diameter through the fringe pattern, with fixed optical path separation and θ steadily varying. Sets of fringes are shown for different values of the phase $\varepsilon_c(<2\pi)$ at the centre of the fringe pattern. It is worth noting the relationship between some of these quantities; at the exact centre of the pattern

$$\varphi_c = 2\pi 2\mu d/\lambda_0 = 2\pi n + \varepsilon_c. \tag{3.41}$$

At an angle of incidence θ the phase is given, with the small-angle approximation for $\cos\theta$, by

$$\varphi = 2\pi 2\frac{\mu d}{\lambda_0}(1 - \theta^2/2)$$
$$= \theta_c(1 - \theta^2/2). \tag{3.43}$$

For successive bright rings, going outward from the centre, $\varphi_1 = 2\pi n$, $\varphi_2 = 2\pi(n-1)$ and $\varphi_3 = 2\pi(n-2)$ so that, on substituting:

$$\theta_1^2 = 2\varphi_c^{-1}(\varphi_c - \varphi_1) = \frac{2\varepsilon_c}{(2\pi n + \varepsilon_c)} \tag{3.43a}$$

$$\theta_2^2 = 2\varphi_c^{-1}(\varphi_c - \varphi_2) = \frac{2(\varepsilon_c + 2\pi)}{(2\pi n + \varepsilon_c)} \tag{3.43b}$$

$$\theta_3^2 = 2\varphi_c^{-1}(\varphi_c - \varphi_3) = \frac{2(\varepsilon_c + 4\pi)}{(2\pi n + \varepsilon_c)} \tag{3.43c}$$

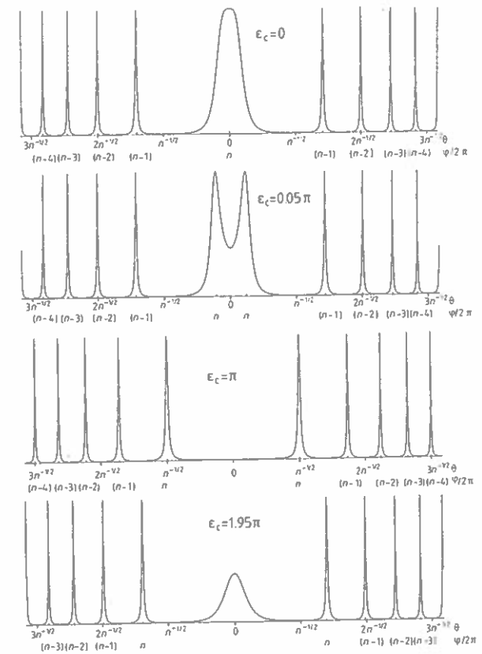

Figure 3.6 Fringe profiles for a uniform change of θ and $F = 256$ and for various values of the phase ε_c at the centre of the pattern. The relative values of the fringe order number $\varphi/2\pi$ are shown; values of θ are expressed in units of $n^{-1/2}$ (equal to $(\lambda/2d)^{1/2}$) which is valid for the small-angle approximation. Finesse $N_R \simeq 25$.

where $\theta_1, \theta_2, \ldots$ are the respective angular radii in an obvious notation. These considerations show that the differences of squares of successive angular radii are equal and of magnitude $4\pi/(2\pi n + \varepsilon_c)$. For a high order of interference (n large) this becomes $2/n$; correspondingly the difference of squares of successive radii is given by $2f^2/n$ where f is the focal length of the fringe forming lens. These quadratic relationships will be returned to frequently in succeeding sections.

It is worth noting other mathematical formulations that have been given for the fringes. The amplitude of the transmitted beams may be written from equation (3.2) in simplified form as

$$A_t(m) = T[1 + Re^{i\varphi} + R^2e^{i2\varphi} + \ldots]. \tag{3.44}$$

Multiplying this series by the complex conjugate gives the transmitted intensity

coefficient as

$$I_t = T^2[1 + R^2 + R^4 + R^6 + \ldots] + T^2 2\cos\varphi[R + R^3 + R^5 + \ldots] + T^2 2\cos 2\varphi[R^2 + R^4 + R^6 + \ldots] + \ldots \quad (3.45)$$

$$= [T^2/(1 - R^2)]\left(1 + 2\sum_{m=1}^{\infty} R^m \cos m\varphi\right) \quad (3.46a)$$

$$= [T^2/(1 - R^2)]\sum_{m=-\infty}^{m=+\infty} R^{|m|} e^{im\varphi} \quad (3.46b)$$

$$= [T^2/(1 - R)^2][(1 - R)/(1 + R)]\left(1 + 2\sum_{m=1}^{\infty} R^m \cos m\varphi\right). \quad (3.46c)$$

This form was employed by Krebs and Sauer (1953) and has some convenience for convolution with Lorentzian and Gaussian functions as discussed further in Appendix 6 (see also §3.7).

A form in which the fringes are represented as a non-periodic function folded with $\sqcap\!\sqcap\!\sqcap$ the Dirac comb was given by Connes in 1961 (see also Steel 1967). Transformation of equation (3.46) to obtain the transfer function followed by inverse transformation gives the expression

$$I_t = \sqcap\!\sqcap\!\sqcap(\varphi/2\pi) * [T^2/(1 - R^2)]\{2\rho/[\rho^2 + (\varphi - 2n\pi)^2]\} \quad (3.47)$$

where $\rho = -\log_e R$, and $\sqcap\!\sqcap\!\sqcap(\varphi/2\pi)$ is the Dirac comb made up of a sequence of delta functions spaced at integral values of $\varphi/2\pi$. The convolution (indicated by symbol $*$) is thus with the non-periodic Lorentzian function $[T^2/(1 - R^2)][2\rho/(\rho^2 + \Delta\varphi^2)]$ where at the peak of the nth-order fringe $\Delta\varphi = (\varphi - 2n\pi) = 0$. In this formulation the half-width of the non-periodic fringe at half the peak intensity is given by

$$|\Delta\varphi_{1/2}| = \rho = -\log R. \quad (3.48)$$

For high reflectivity R this differs from the half-width of equation (3.36) by terms of the order of $(1 - R)^3$. The fringes are thus considered as a succession of overlapping Lorentzian functions. This is not to say that a single complete fringe extending over $-\pi \leqslant \Delta\varphi \leqslant \pi$ may be considered as of Lorentzian profile. It may be so approximated only after correction for the overlap (which is relatively the largest in mid-order) of the other (Lorentzian) fringes. For R sufficiently large the sequence of Lorentzian functions may be derived more directly from equation (3.29) and is given by

$$I_t = [T^2/(1 - R)^2]\sum_{n=0}^{\infty}\{1 + F[(\varphi - 2n\pi)/2]^2\}^{-1}. \quad (3.49)$$

Precision analysis of instrumental profiles in terms of a succession of Lorentzian profiles was apparently first employed by Vaughan (1962).

3.2.4 Response to various light sources

The earlier discussions establish that an interferometer of fixed spacing is a very

sensitive angle-dependent filter. For any particular angle of incidence the transmission is determined by equations (3.29) and (3.31) so that the interferometer, considered as a filter, may be finely tuned by varying the optical separation of the plates. In this regard it is worth considering the response of the system to both different types of light source and different optical arrangements that pass the light into the interferometer.

The light source is usually set at the focus of a collimating lens. Light from a single point in the source is thus introduced as a parallel beam into the interferometer. The fringe forming lens forms in its focal plane an image of the source; the illumination in the image is modulated by the interference. For a large-'area' source several bright rings will be observed on the image plane. For a narrow 'filament' or 'streak' source (such as the scattering at right angles to a laser beam, or a gas discharge column) the narrow image on the fringe plane will illuminate a section of the fringe. As indicated in figure 3.4 this section will only pass through the centre of the fringe pattern if the source intersects the optic axis determined by the collimating lens and the normal to the optical surfaces of the interferometer. Similarly for a point or small-area source (e.g. light scattering from a narrow sample, or a small hollow cathode) the illumination in the fringe plane will depend on where its image falls in relation to the 'fringes' determined by the interferometer and fringe forming lens. If the source lies on the optic axis as defined above it will automatically illuminate the centre of the fringe pattern. In general precise optical adjustment will be needed to ensure this.

It may at first sight seem rather puzzling as to why sharp fringes are still observed if the collimating lens is defocused or even removed altogether. Figure 3.7 shows an interferometer illuminated by a source without a collimating lens. Consider light from a point a′ passing into the interferometer at an angle of incidence θ. If at this angle the usual constructive interference condition is obeyed, multiple interferences will add to give a bright region in the focal plane at a″ as shown. This region will obviously lie on the bright circular fringe due to other rays of equal incidence from the same point a′ in the source. Now consider light from another point b′ in the source plane; a ray from b′ parallel to the original ray from a′ has the same constructive interference condition in the interferometer and due to the lens action is similarly brought to point a″ in the focal plane as shown. These considerations similarly apply to all points of the source, and the illumination at a point in the fringe is thus an incoherent summation of contributions from all over the source. In the focal plane the form of the fringes is unchanged whether the incoming light is collimated or not; the interferometer is sometimes said to 'make its own parallel light'.

What then is the practical function of the collimating lens? Most obviously it serves to increase the illumination in the fringe plane. This is easily appreciated by supposing initially that the interferometer is removed. With a broad or area source, considerable departure from precise collimation leads only to a marginal enlargement of the defocused image of the source at the fringe plane and a small reduction of illumination. Similar considerations apply if the source has considerable depth along the optic axis—precise collimation can only be

achieved in any case for one particular plane normal to the axis, usually chosen in the middle of the source. Under these conditions it is clear that the collimating lens need not be a particularly high quality optical component. On the other hand for small sources, either narrow or point-like, any lack of collimation ensures a badly defocused image, loss of illumination and a considerable reduction of peak intensity in the fringes.

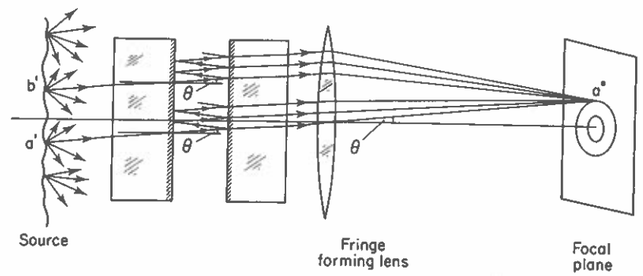

Figure 3.7 'The interferometer makes its own parallel light', showing the formation of sharp fringes with an uncollimated source. The interferometer acts as an angular filter and fringes of equal inclination are formed in the focal plane with light contributed (incoherently) from all over the source. A complete ring system is indeed created with a single point source (compare figure 3.4).

3.3 Line width, finesse and resolving power

3.3.1 Fringe width and finesse

From the earlier discussions it is seen that narrow fringes are formed in transmission. The sharpness of the fringes increases with increasing reflectivity (figures 3.5, 3.6) and this may be conveniently expressed as the ratio of the separation of successive fringes to the fringe width at half-intensity. (This is often expressed as the 'full width at half-height' and abbreviated to FWHH, but is also unfortunately sometimes called the 'half-width'. This latter term is preferably reserved for the 'half-width at half-height', (HWHH) or at least extended to 'half-height width'. In many papers and texts only careful reading can establish which meaning is being used.) For quasi-monochromatic light the positions of the half-intensity points relative to a fringe of order n are given by

$$\varphi = 2n\pi \pm (\varphi_{1/2}) \tag{3.50}$$

where the periodic part of equation (3.29) defines $\varphi_{1/2}$ such that

$$[1 + F\sin^2(\varphi_{1/2}/2)]^{-1} = \tfrac{1}{2}. \tag{3.51}$$

From the discussion following equations (3.35)–(3.37), the ratio of separation to

full width at half-height, equal to $\pi/\varphi_{1/2}$, is called the finesse N_R and is given by

$$N_R = \pi/\varphi_{1/2} = (\pi/2)\sqrt{F} = \pi\sqrt{R}/(1 - R). \tag{3.52}$$

The quantity $\pi\sqrt{R}/(1 - R)$ is sometimes called the 'reflectivity finesse' as implying that this is what would be found for ideally flat étalon plates with that reflectivity. In practice the instrumental function and linewidth will be modified by plate imperfections and the manner in which the instrument is used—for example by aperture broadening. These effects are discussed in §§3.4 and 3.6.

3.3.2 Free spectral range

The discussion so far has concentrated on an interferometer illuminated with quasi-monochromatic light of wavelength λ_0. The interference fringes are governed by the phase delay φ between successive beams given by equation (3.1). For a slightly different wavelength $\lambda_0 + \delta\lambda$ another system of interference fringes is similarly formed. At a fixed angle of incidence θ the *difference* of phase delays for the two wavelengths is given by

$$|\delta\varphi| = 2\pi(2\mu d\cos\theta)\delta\lambda/\lambda_0^2 \tag{3.53}$$

supposing that any change of refractive index between the two wavelengths is negligible. The expression $\delta\lambda/\lambda_0^2$ is of course the wavenumber difference between the two wavelengths.

If $|\delta\varphi|$ is equal to 2π the sets of fringes overlap exactly so that the nth order of interference of wavelength λ_0 matches the $(n - 1)$th order of interference of $\lambda_0 + \delta\lambda$. In consequence from equation (3.53)

$$2\pi = 2\pi(2\mu d\cos\theta)[\delta\lambda/\lambda_0^2]_{2\pi} \tag{3.54}$$

and

$$[\delta\lambda/\lambda_0^2]_{2\pi} = (2\mu d\cos\theta)^{-1}. \tag{3.55}$$

The wavenumber interval corresponding to the right-hand side of equation (3.55) (with $\cos\theta$ set equal to 1) is usually called the 'free spectral range' of the interferometer and assigned the symbol σ_1 or σ_d. It is thus the wavenumber difference between two spectral lines which would overlap exactly in successive orders for the given plate separation.

The significance of the free spectral range becomes clear when one considers the term $|\delta\varphi|$ in equation (3.53). The wavenumber difference $\delta\sigma$ between two close lines is given by

$$\delta\sigma = \delta\lambda/\lambda_0^2 = \frac{|\delta\varphi|}{2\pi}(2\mu d\cos\theta)^{-1} \tag{3.56}$$

$$= \frac{|\delta\varphi|}{2\pi}\sigma_1. \tag{3.57}$$

Thus if the interval between the two lines can be measured (as given by $|\delta\varphi|/2\pi$) as a fraction of an order, the wavenumber interval can be determined. Details of this procedure are discussed later in this section. One needs initially to examine the implications for the resolving power of the instrument.

3.3.3 Resolving power

As $|\delta\varphi|$ becomes smaller the fringe systems of the two lines become closer and will eventually begin to overlap. As discussed in §2.3 the Rayleigh criterion for resolution was derived for prism and grating instruments on the basis of diffraction; the intensity mid-way between the peaks turns out to be 0.811 of the peak intensities. Such a saddle-to-peak ratio could be adopted for the interferometer as a criterion for resolution and the corresponding resolving power determined. Nowadays such a procedure would seem somewhat artifical, particularly when one considers that the instrumental profile, unlike that for prism and gratings, is without subsidiary maxima and minima. It is in fact much more straightforward to see what happens if $|\delta\varphi|$ in equation (3.53) is set equal to $\varphi_{1/2}$ in equation (3.50). The two fringe systems then coincide at their half-intensity positions; for components of equal intensity, and for large F the ratio of saddle-to-peak is 1 : 1.2, almost identical to the Rayleigh criterion.

Thus as a matter of convenient usage two equal lines may be considered as resolved if

$$|\delta\varphi|_{\min} = \varphi_{1/2} = 4/\sqrt{F} = 2\pi/N_R. \tag{3.58}$$

With this criterion the minimum wavenumber separation which may be resolved (the 'resolving limit') is, from equation (3.57), given by

$$\Delta\sigma_{\min} = \sigma_1/N_R \tag{3.59}$$

i.e. by the free spectral range divided by the finesse.

In equation (3.53) the term $(2\mu d\cos\theta)/\lambda_0$ may be set equal to n for high order of interference and thus $|\delta\varphi|$ may be written as

$$|\delta\varphi| \sim 2\pi n\delta\lambda/\lambda_0. \tag{3.60}$$

In consequence the limiting resolving power is given by

$$\mathcal{R}_0 = \lambda_0/\delta\lambda = 2\pi n|\delta\varphi|^{-1}_{\min} \tag{3.61}$$

$$= 2\pi n(N_R/2\pi) = nN_R. \tag{3.62}$$

(It may be noted that if the exact Rayleigh criterion is adopted the resolving power becomes $0.97nN_R$.)

This result has a well known universal character and expresses the formulation that the resolving power of an optical system is given by the number of wavelengths within the optical path difference of the extreme rays. For a grating instrument the resolving power is given by nN_g where N_g is the number of interfering beams due to the number N_g of ruled elements and n for the first, second, third, . . . order of interference is of course 1, 2, 3 etc. In equation (3.1) for the interferometer the order of interference n is much larger and given by $2\mu d/\lambda_0$—the number of wavelengths within the optical separation of the plates; by analogy with the grating, N_R is sometimes called the 'effective number of interfering beams' (see also §2.3).

It must be emphasized that the resolving power $\mathcal{R}_0$ and resolving limit $\Delta\sigma_{\min}$ just calculated represent limiting values for perfect plates with vanishingly small

detector aperture, grain size etc. In practice they are degraded by many factors including plate imperfections and finite aperture; in spectrometric use an effective choice of aperture must be made that makes a compromise between increasing light flux and inevitable loss of resolving power. This is discussed in detail in §3.4 and Appendices 8 and 10.

3.3.4 Overlap of orders

The analysis of spectra may be complicated by the overlap of orders of different spectral features. This problem is illustrated in figure 3.8 which shows sequences of fringes for two spectral lines A and B. As labelled in the figure the difference of phase delays between the two lines is

$$\Delta\varphi = 2\pi m - (2\pi n - \varepsilon')$$
$$= 2\pi(m-n) + \varepsilon'. \tag{3.63}$$

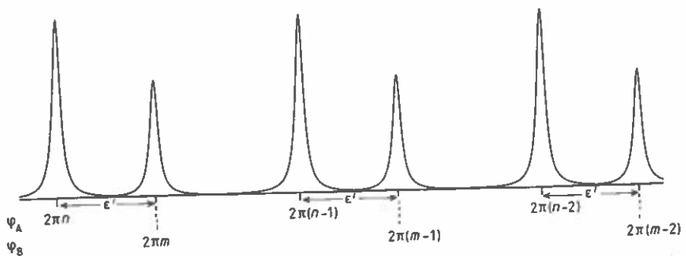

Figure 3.8 Two fringe systems of wavelengths λ_A and λ_B are shown with their respective orders of interference and an observed phase interval ε' or fraction of an order $\varepsilon'/2\pi$. Reduction of order number to the right-hand side corresponds to a decrease in the optical path separation.

The order number difference is thus $(m-n) + \varepsilon'/2\pi$ and the wavenumber interval

$$\Delta\sigma = \sigma_1[(m-n) + \varepsilon'/2\pi] \tag{3.64}$$

where σ_1 is the free spectral range of the interferometer. The problem is that, while in practice the fraction of an order $\varepsilon'/2\pi$ may readily be measured from the spectral record (for which theory and practice are outlined in succeeding sections), the difference of the order numbers m and n is unknown. This gives rise to the following possibilities for the wavenumber interval (where the sign or direction of separation has been ignored for the moment):

$$|\Delta\sigma| = \sigma_1\varepsilon'/2\pi,\ \sigma_1(1 + \varepsilon'/2\pi),\ \sigma_1(2 + \varepsilon'/2\pi)\ \ldots \tag{3.65a}$$

$$|\Delta\sigma| = \sigma_1(1 - \varepsilon'/2\pi),\ \sigma_1(2 - \varepsilon'/2\pi),\ \sigma_1(3 - \varepsilon'/2\pi)\ \ldots. \tag{3.65b}$$

In general there is no way of deciding which of these is correct other than by

making another measurement with a different separation of the interferometer plates and a different free spectral range σ_2. A second set of possible shifts may then be written out as

$$|\Delta\sigma| = \sigma_2\varepsilon''/2\pi,\ \sigma_2(1 \pm \varepsilon''/2\pi),\ \sigma_2(2 \pm \varepsilon''/2\pi)\ \ldots \tag{3.66}$$

By comparison of the two sets it should be possible to determine uniquely the actual wavenumber separation. Very occasionally it may happen that, due to experimental uncertainties in the measurement of ε' and ε'', more than one possibility may remain; a third measurement may be necessary. However in many cases good *a priori* knowledge may exist (e.g. previous measurements, expected ratio of line width and separation, known range of Brillouin shift for the class of materials etc) that allows immediate assignment of the basic order number difference on the first measurement alone.

Having determined the wavenumber difference one needs to establish the sign—which line has the higher wavenumber or shorter wavelength. For this one needs to know the direction of increasing or decreasing order number. As drawn in figure 3.8 the order number is decreasing towards the right-hand side. This result would be deduced in practice if it corresponded to a reduction of the optical path separation, or equivalently to moving out from the centre of the fringe pattern. Thus, returning to first principles, at the position of the left-hand peak of A one may write the optical path separation as

$$n\lambda_A = (m + \varepsilon'/2\pi)\lambda_B. \tag{3.67}$$

From the measured shift the difference $(m-n)$ is determined. Suppose that the measured shift is in fact $\sigma_1(1 - \varepsilon'/2\pi)$ from the range of possible values in equation (3.65). Then $m-n = -1$ so that n is greater than $(m + \varepsilon'/2\pi)$; in consequence $\lambda_A < \lambda_B$ and $\Delta\sigma_{A-B}$ is negative. The important point is that one must determine the direction of increasing or decreasing order number. As discussed in the following section, on a photographic recording this is immediately obvious; with spectrometric recording one needs to know whether the optical path separation is increasing or decreasing.

3.4 The Fabry–Perot spectrometer

With direct photoelectric recording of light in the fringes the Fabry–Perot interferometer may be used as a spectrometer. This technique was first introduced by Jacquinot and Dufour in 1948 and is now extensively applied. An aperture defining a specified wavenumber interval is placed in the focal plane of the fringes and the light passing through the aperture is then measured on a detector. In principle several techniques can be employed for scanning over the required spectrum—for example the aperture may be moved in the fringe plane thus examining different wavenumber intervals, or a series of apertures may be used each with its own individual detector. Such methods are in fact difficult but arrangements of this type, developed for pressing experimental requirements

and where the light source permits, are described in the next chapter. In practice far and away the most common, and certainly the simplest technique, is to employ a fixed aperture and then, by suitable variation of the optical path between the étalon plates, scan the fringes across the aperture. Apertures of various form may be employed, but in practice only two—annular rings, or much more commonly, a single circular pinhole aperture, centred on the fringe pattern—have been widely used. The choice of aperture size and its effect on optical resolution and light gathering needs to be considered.

3.4.1 *Aperture size and resolving power*

The wavenumber interval embraced by a circular aperture placed at the centre of a fringe pattern is readily calculated. In figure 3.4 consider first a vanishingly small aperture on axis at the centre of the fringe pattern in the focal plane. As the optical path between the interferometer plates is steadily reduced, successive fringes are observed shrinking down onto the axis. For monochromatic light (and ideally flat plates) a detector would record the undistorted fringe profiles as given in equation (3.8) and figure 3.5. Fringe maxima are observed as the optical separation of the plates becomes equal to integral values of the wavelength (i.e. $2\mu d = n\lambda_0$). The apparent free spectral range or spectral interval between successive orders (see §3.3.2) is equal to $(2\mu d)^{-1}$. Now consider the pinhole increased in size such that it embraces a non-vanishing wavenumber interval. Suppose that at the centre of the fringe pattern for a given optical path separation μd the order of interference is $(n + \varepsilon_c/2\pi)$ and that at the periphery of the aperture the order is $(n + \varepsilon_c/2\pi - k/2\pi)$. The spectral interval which is covered by such an aperture is thus a fraction $(k/2\pi)$ of an order equivalent to $(k/2\pi)(2\mu d)^{-1}$ wavenumber. If the angular radius of the aperture (relative to the fringe forming lens) is θ_k the expressions governing these terms are

$$2\mu d = (n + \varepsilon_c/2\pi)\lambda_0 \tag{3.68a}$$

and

$$2\mu d\cos\theta_k = (n + \varepsilon_c/2\pi - k/2\pi)\lambda_0. \tag{3.68b}$$

For small θ this becomes

$$2\mu d(1 - \theta_k^2/2) = (n + \varepsilon_c/2\pi - k/2\pi)\lambda_0 \tag{3.69}$$

and from these

$$\mu d\theta_k^2 = (k/2\pi)\lambda_0 \tag{3.70}$$

in consequence

$$k/2\pi = \mu d\theta_k^2/\lambda_0 \tag{3.71}$$

and

$$\theta_k = [(k\lambda_0/2\pi)/\mu d]^{1/2}. \tag{3.72}$$

If the aperture is of diameter a, and the focal length of the fringe forming lens is f, since $\theta_k = a/2f$ the fraction of an order observed by the aperture is

$$k/2\pi = (\mu d/\lambda_0)(a/2f)^2 \tag{3.73}$$

and the wavenumber interval $\Delta\sigma_k$ is

$$\Delta\sigma_k = (k/2\pi)(2\mu d)^{-1} = (1/2\lambda_0)(a/2f)^2. \tag{3.74}$$

It is worth appreciating that each wavenumber increment within this interval is given equal weighting. This arises because the wavenumber increment corresponding to an increase in aperture diameter δa is proportional to a, as is the corresponding increment in area. Quite simply the increment in wavenumber is given by

$$\delta(\Delta\sigma) = \lambda_0^{-1}(a/4f^2)\delta a \tag{3.75}$$

and the light flux collected in this increment is proportional to the area of the annulus which is

$$\delta_{\text{Area}} = (\pi/2)a\delta a. \tag{3.76}$$

The flux per unit wavenumber interval is thus constant. In consequence an accurately centred pinhole provides an instrumental function of rectangular or 'top hat' form. The way in which this contributes to the shape of an observed spectrum is discussed in the final section of this chapter.

It is worth remarking that the peak of the fringe profile is now displaced slightly from the condition $2\mu d = n\lambda_0$. Simple considerations of scanning a rectangular function over a symmetrical spectral feature of centre wavelength λ show that the maximum transmission will occur when the rectangular function is itself set symmetrically over the peak. The recorded fringe is thus displaced a distance $k/4\pi$ of an order from that which would be found with a vanishingly small aperture. For symmetrical, well separated spectral features this is of little account, all are affected similarly. In compound spectra e.g. isotope shifts, hyperfine structure, this may not be true and correction to the measured separations or shifts will have to be applied.

For a given separation of the interferometer plates it is often convenient to calculate, as a number of easy experimental reference, the diameter $a_{2\pi}$ of the 'first full ring'. This is the diameter of the first fringe from the centre supposing that a bright fringe exists exactly at the centre of the pattern (i.e. $\varepsilon_c = 0$ and $k = 2\pi$ in equations (3.68)). Substituting in these equations and taking $\mu \sim 1$ for air-spaced plates we have

$$a_{2\pi} = 2f\theta_1 = 2f\sqrt{\lambda/d}. \tag{3.77}$$

The choice a of any physical aperture may then be considered immediately on the basis that it will effectively embrace a fraction $(a/a_{2\pi})^2$ of the free spectral range. Thus for $\lambda = 0.5\ \mu\text{m}$, $d = 5$ mm and $f = 10$ cm the diameter of the first full ring is 2 mm; a 200 μm pinhole is immediately seen to transmit $\frac{1}{100}$th of an order or a wavenumber interval of $10^{-2}\ \text{cm}^{-1}$.

3.4.2 Light collection

As shown in §2.3 the light gathering power of an optical system is characterized by the 'étendue' U, the product of source area observed and the collection solid angle. From figure 3.9, showing the basic geometry of a spectrometer, the étendue is given by

$$\begin{aligned} U &= (\pi/4)(af_1/f)^2(\pi/4)(D/f_1)^2 \\ &= (\pi D/4)^2(a/f)^2. \end{aligned} \tag{3.78}$$

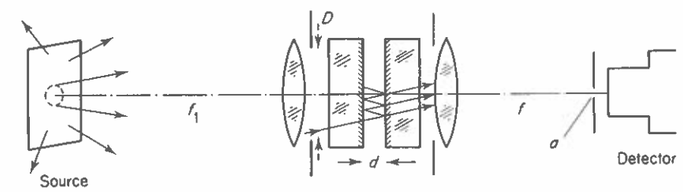

Figure 3.9 The geometry of a basic Fabry–Perot spectrometer showing the central scanning aperture of diameter a, usable plate diameter D and the area source. When correctly focused, light is detected only from within the dotted circle at the source of diameter (af_1/f).

This establishes the result that with a pinhole aperture collecting over a fixed wavenumber interval, for which a/f is constant (equation (3.74)), the étendue is independent of lens geometry and can only be increased by making D larger—that is by employing larger étalon plates. (It is not infrequently supposed that by reducing f_1, and thus increasing the collection solid angle, that light gathering can be increased. Clearly this is a misconception.) The light collected is still of course proportional to source brightness. Nevertheless these considerations of étendue do not rule out an optimum choice of pinhole and lens geometry but this is governed by several other experimental factors which include: (i) the line widths and positions of the spectral components under study; (ii) the f number or angular aperture of any auxiliary dispersing equipment and its dispersive power and (iii) the nature of the light source and what degree of choice or flexibility is available in its size, brightness, working distance etc.

Considering these factors in turn it might be supposed at first sight that the scanning aperture should be chosen to correspond to some (small) fraction of the instrumental line width determined by the reflectivity finesse N_R, that is some fraction of σ_1/N_R. It is well known, and indeed easily verified, that a rectangular function folded with (or equivalently scanned over) a smooth profile more than about five times its width introduces very little distortion. For profiles of Gaussian or Lorentzian form the rectangular function may be up to one third or even one half of their width (at half-intensity) without gross distortion of their shape; the alterations which do occur are most pronounced at the peak and

in regions where the shape of the curve is changing most rapidly†. These considerations might suggest that with a Fabry–Perot spectrometer of reflectivity limited finesse of say 50 a pinhole aperture corresponding to less than $\sim \frac{1}{150}$th of an order should be chosen. In practice this may not make best use of available light. It is likely that the spectral features themselves will be considerably wider than the finesse limited instrumental width (see e.g. the evaluation of Hernandez (1982a) and Appendix 10). Continuing the numerical illustration, with spectral features of width say $\frac{1}{10}$th of the free spectral range, the pinhole aperture might be chosen up to $\frac{1}{30}$th of an order. This increase of up to a factor of five in light collection compared with the figure above could be most valuable for low-brightness sources. For a spectral line having many spectral features of different separation, intensity and shape, detailed consideration will need to be given, supported if necessary with computed simulation, to the choice of aperture which provides an optimum between light collection and freedom from spectral distortion. The analysis and possible techniques of deconvolution of spectra are discussed later in §3.7.

3.4.3 Auxiliary dispersion and optical train

The second factor arises from any requirements for auxiliary dispersion. In many situations, e.g. studies of scattered laser light or well separated atomic spectra, simple optical filters may be adequate. However in many cases an auxiliary spectrometer will be required. Several arrangements employing prism and grating instruments for photographic applications have been described by Tolansky (1947, chapter 9). The present problem is illustrated with reference to the Littrow mounted instrument shown in figure 3.10.

Consider the light under study of wavelength λ_0, with a near neighbour λ_1 which has to be discriminated against in the recording. It is supposed that the primary aperture at A is placed in the focal plane of the lens L_s so that the reflection grating (which could equally well be a dispersing prism and mirror) is illuminated with parallel light. A dispersed image of the same size as the aperture is thus formed in the plane B. A complementary aperture of size a may accordingly be positioned at B and only light passing through this will be recorded. For a source of brightness $\mathcal{B}$ and a combined transmission of interferometer and grating $\mathcal{T}$ the flux collected is given by

$$\mathcal{F} = (\pi/4)^2(a/f)^2 D^2 \mathcal{B}\mathcal{T}. \tag{3.79}$$

†In Appendix 8 the resolving power $\mathcal{R}$ and luminosity $\mathcal{L}$ of a Fabry–Perot spectrometer for monochromatic light are discussed on the lines first considered by Jacquinot and Dufour (1948). It turns out that a broad maximum of the product $\mathcal{R}\mathcal{L}$ is found for a pinhole aperture corresponding to $\sim N_R^{\frac{1}{2}}$ of an order. This provides a useful comparison of étendue with e.g. the confocal interferometer (Chapter 5). However the resolving power is considerably reduced to $\mathcal{R} \sim \mathcal{R}_0/\sqrt{2}$. In certain cases this may not be tolerable, particularly where precision analysis of line profiles is required or the resolution of close components of differing intensity. In practice the choice of aperture should be dictated by the nature of the spectral features under study, the information required, and the available light flux (see also Appendix 10).

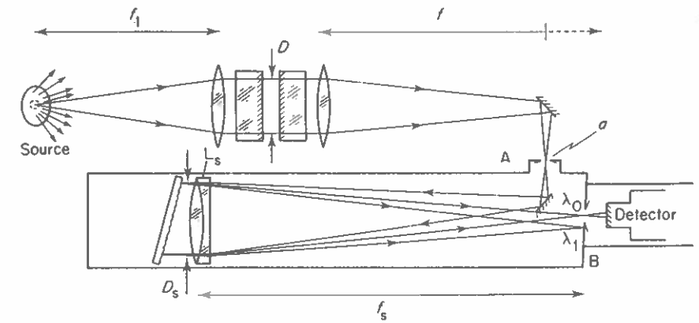

Figure 3.10 A Fabry–Perot spectrometer with auxiliary, Littrow mounted, grating spectrometer, shown with two wavelengths λ_0 and λ_1. In practice the limiting parameters are likely to be the effective size of the grating D_s and its dispersive power. These determine the light gathering power of the complete instrument (see the text).

In practice the limiting component is likely to be the effective diameter D_s of the available grating. Thus the usable diameter D of the étalon plates is given by

$$D/f \leqslant D_s/f_s. \tag{3.80}$$

If the dispersive power of the grating is given by α radians per unit wavelength interval, the separation of the two lines in the focal plane B is given by

$$a_s = \alpha|\lambda_0 - \lambda_1|f_s \tag{3.81}$$

and thus the maximum allowable size for the aperture is $a \leqslant a_s$. Inserting these terms, the maximum light collection is thus given by

$$\begin{aligned}\mathcal{F} &\leqslant \left(\frac{\pi}{4}\right)^2\left(\frac{\alpha|\lambda_0 - \lambda_1|f_s}{f}\right)^2\left(\frac{D_s}{f_s}f\right)^2\mathcal{B}\mathcal{T}\\ &\leqslant \left(\frac{\pi}{4}\right)^2(\alpha|\lambda_0 - \lambda_1|D_s)^2\mathcal{B}\mathcal{T}.\end{aligned} \tag{3.82}$$

Most interestingly the light collection can only be increased by use of a larger or more dispersive grating. The experimentalist has clearly to plan his photon and financial budget on this basis! The upper limit indicated in equation (3.82) will of course only be approached with the proper choice of a and D in relation to the available lenses f, f_1 and f_s, and the considerations of the scanning interval described earlier in this section.

The final point is concerned with the degree of flexibility or choice in the light source itself. Underlying the preceding discussion of étendue and light collection has been the supposition that the source is at least of cross sectional dimension

$a(f_1/f)^2$. It does not however require to be a great deal bigger; light coming from outside this circular zone is not passed directly to the detector and indeed may be a positive disadvantage if, on account of parasitic scattering within the optical train, it contributes to a broad incoherent background in the recorded spectra. It is worth noting that in principle, for large-area sources, the light collected within a central circular aperture (of collection interval a fraction k of an order) could be augmented by successive annular apertures extending from the first, second, . . . full bright rings for a similar fraction k of an order. The area of each of these annuli would of course be the same as that of the central circular aperture, so that factors of 2, 3, . . . in light collection would be attained. In practice construction of such a compound aperture presents considerable difficulties and its inflexibility is a considerable disadvantage compared with a simple circular aperture†.

In general one is thus left with increasing the source brightness and ensuring that the aperture size is matched most appropriately to the source size. Relevant considerations appropriate to particular experimental fields are discussed in Chapters 7–10. It is however worth remarking that practical limits on source brightness in the case of atomic spectra may be set by phenomena such as self-absorption and electric field and Doppler broadening. In laser scattering studies limitations may be imposed by heating and sample damage, and on collection angle by the allowed spread of scattering vectors.

3.5 Photographic recording and analysis of fringes

Some basics of photographic recording are outlined in §2.4.1. The design, construction and operation of static Fabry–Perot interferometers for photographic use are described in §4.3. As the earliest means of recording spectra, the production and analysis of photographic fringes has been discussed in detail in several early reviews (e.g. Meissner 1941, Tolansky 1947). The present account attempts to select some of the more important features.

3.5.1 Photographic recording

In order to record Fabry–Perot fringes the photographic plate must either be

†Nevertheless renewed interest has recently been shown in the potential of the technique. Dupoisot *et al* (1978) and Dupoisot and Prat (1979) calculated the gain in flux for a grid aperture made up of a photographic positive of the Fabry–Perot rings. Such a grid obviously provides a continuous variation of transparency. Hernandez *et al* (1981) described a twin étalon scanning spectrometer (TESS) in which the transmission characteristics of the grid are replaced by those of the second fixed étalon. Both fixed and scanning étalons require to have the same spacing. The instrumental function is the convolution of the two individual profiles; Hernandez *et al* showed that the form is equivalent to that of a single étalon with an effective reflectivity $R_e = R_1R_2$. Using the TESS spectrometer with 21 fringes they demonstrated a gain in luminosity of 13.6, in good agreement with calculation. A detailed analytical description of the TESS spectrometer has been given by Hernandez (1982b).

placed in the focal plane of the fringe forming lens (e.g. as in figure 3.4) or in the conjugate plane of auxiliary dispersing equipment (e.g. as in figure 3.10). In the latter case the circular aperture would be replaced by a vertical slit (normal to the plane of the diagram) of width determined by the dispersion. To allow for the change of focal length of L_s with wavelength the photographic plate at B must be tiltable; simultaneous recording of a wide band of wavelengths may require a flexible plate bent in an appropriate curve. It is most important that the optical components following the interferometer are of very high quality; rays across the full area of the étalon plates must be brought to a precise focus and, in photographic use particularly, over a full range of off-axis fringes. The initial setting-up procedure will usually be carried out with a series of test exposures at different lens settings using a narrow-band source (such as a cadmium, mercury isotope or neon lamp). Figure 3.11 shows fine fringes in a neon spectrum due to Meissner (1941) taken with a 10 mm étalon.

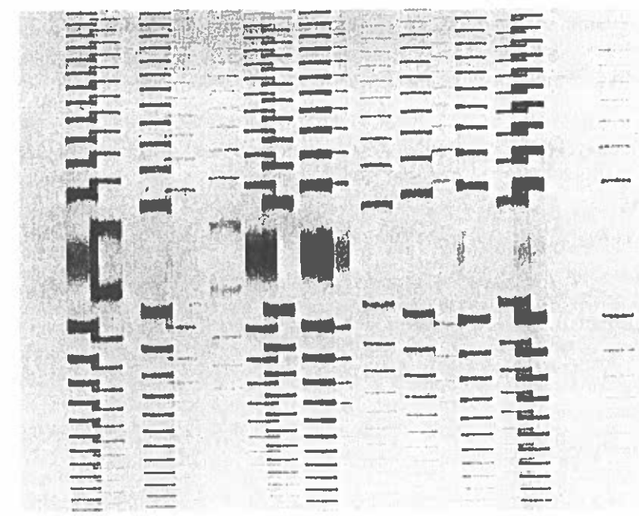

Figure 3.11 A photographic recording (negative) of sharp Fabry–Perot fringes in neon, taken with an interferometer of 10 mm spacing (from Meissner 1942). Reproduced by permission of the Optical Society of America.

Once the equipment has been satisfactorily set up, trial exposures on the experimental source may be made. It is important at this stage that the appropriate conditions of use are established—for example is the source itself focused onto the detector plane (as in figure 3.4) or should light be integrated from the source without focusing (figure 3.7)—see also the discussion in §3.2.4.

It may be important for a variety of reasons (e.g. high electric fields) to exclude certain areas of the source by suitable masking. The relative intensity of components in the source is of paramount importance. Given narrow, symmetric lines of approximately equal brightness, single, suitably chosen, exposures should suffice to produce recordings from which the peaks of the fringes may be accurately measured and compared. Usually this will not be the case, and it may not prove possible to make a single exposure in which all the components lie within the linear portion of the characteristic curve shown in figure 2.6. This is well illustrated in figure 3.12 due to Kuhn and Series (1950) showing three photometer tracings of the Hα line for short, standard and long exposures. In this instance the following experimental technique was developed: three successive exposures were made on one plate with exposure times in the ratio of 1 : 10 : 1. The positions of the strong fringes (1) and (2) were measured from a

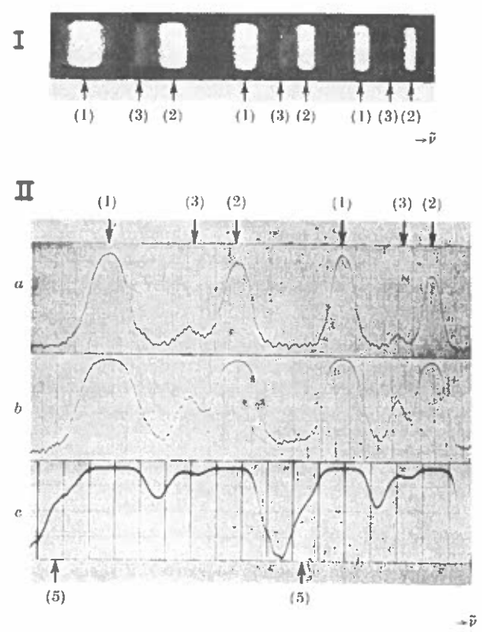

Figure 3.12 Fine structure in Hα recorded with a Fabry–Perot interferometer of 7.67 mm spacing showing one side of the circular fringe pattern: I, a photographic positive of three orders of interference; II, photometer tracings of two orders of interference at different exposure times (*a*, short; *b*, standard; *c*, long). Note the asymmetry due to the weak unresolved component at (5) in *c*. (From Kuhn and Series 1950.) Reproduced by permission of The Royal Society.

reference mark on the first and last exposures; the faint fringe (3) was measured from the same reference in the middle exposure. Good agreement of the first and third exposure was an obvious experimental requirement.

With asymmetric fringes, due for example to the overlap of fainter, unresolved components or the wing of strong neighbouring components, reciprocity failure and 'pulling' effects can be a serious problem in assessing the correct peak position. In addition to comparison of measurements made with different exposures, recordings with different interferometer spacing may serve to vary the degree of overlap. In one early example Houston and Hsieh (1934) went so far as to make a series of photographic recordings of the fine structure of the Balmer series at small increments of interferometer spacing. Their aim was to find the precise spacing for which one component was exactly at the mid-order of the other so that 'pulling' effects due to overlap would be negligible. Other possible systematic errors in photographic use, of particular importance for multiple interferometers, are outlined in §6.2.2.

In the analysis of fringes the primary data are obviously present on the photographic plate. Nevertheless photometer recordings often provide a useful and easy appreciation of asymmetric and weak structures, particularly in conjunction with test exposures for plate density and reciprocity. In the past some controversy has existed on the relative merits of measuring the plate directly, or the photometer record. Without first-hand experience the author hesitates to comment; the obvious question would be 'given two measuring techniques, do the results agree—and if not, why not?'

3.5.2 *Analysis of photographic recordings*

The angular radii of successive bright fringes are written down in equations (3.43); the overlap of orders for sets of fringes of neighbouring lines is discussed in §3.3.4 and equations (3.65) and (3.66). Photographic recording has been widely applied to two particular problems which are analyzed in turn: (i) the measurement of the interval between neighbouring components and (ii) the measurement of wavelength by comparison with a standard line.

(a) *Measurement of frequency interval*

Given two close lines A and B separated by a wavenumber interval $\Delta\sigma$, the corresponding fringes (of identical order of interference) will be separated by an order difference δn (which may be a fraction plus a small integer) given by

$$\Delta\sigma = \delta n \sigma_1 = \delta n / 2d_1. \tag{3.83}$$

From equation (3.43) the fringe diameters of the pth and $(p+1)$th bright ring from the centre may be written down for line A as

$${}_aD_p^2 = 4f^2\{2[{}_a\varepsilon_c + (p-1)2\pi]/(2\pi n_a + {}_a\varepsilon_c)\} \tag{3.84a}$$

$${}_aD_{p+1}^2 = 4f^2[2({}_a\varepsilon_c + p2\pi)/(2\pi n_a + {}_a\varepsilon_c)] \tag{3.84b}$$

where f is the focal length of the fringe forming lens. From these equations

$${}_aD_{p+1}^2 - {}_aD_p^2 = 8f^2 2\pi/(2\pi n_a + {}_a\varepsilon_c) \simeq 8f^2/n_a = \Delta_a \tag{3.85}$$

and similarly the fractional order number at the centre ${}_a\varepsilon_c/2\pi$ is given by

$$\begin{aligned} {}_a\varepsilon_c/2\pi &= [{}_aD^2_{p+1}/(8f^2/n_a)] - p \\ &= [{}_aD^2_{p+1}/({}_aD^2_{p+1} - {}_aD^2_p)] - p \\ &= ({}_aD^2_{p+1}/\Delta_a) - p. \end{aligned} \tag{3.86}$$

Similarly for line B

$${}_bD^2_{p+1} - {}_bD^2_p = 8f^2/n_b = \Delta_b \tag{3.87}$$

and

$${}_b\varepsilon_c/2\pi = ({}_bD^2_{p+1}/\Delta_b) - p. \tag{3.88}$$

For closely spaced lines the difference between Δ_a and Δ_b is very small and may be disregarded. The actual order of interference at the centre of the fringe pattern for line A is given by $n_a - {}_a\varepsilon_c/2\pi$ and similarly for line B. Thus the order difference δn is given by

$$\begin{aligned} \delta n &= (n_a - {}_a\varepsilon_c/2\pi) - (n_b - {}_b\varepsilon_c/2\pi) \\ &= (n_a - n_b) - ({}_a\varepsilon_c - {}_b\varepsilon_c)/2\pi. \end{aligned} \tag{3.89}$$

The small integer $(n_a - n_b)$ requires in general to be determined by measurements with a second spacer (§3.3.4); most importantly the fractional order difference $({}_a\varepsilon_c - {}_b\varepsilon_c)/2\pi$ is given by

$$({}_a\varepsilon_c - {}_b\varepsilon_c)/2\pi = ({}_aD^2_{p+1} - {}_bD^2_{p+1})/\Delta \tag{3.90}$$

from which it follows that $({}_aD^2_{p+1} - {}_bD^2_{p+1})$ is a constant which we set equal to δ_{ab}.

Using these formulae δ_{ab} and Δ may be readily derived from photographed spectra, and the spectral interval obtained. If several fringes are recorded Tolansky (1931) suggested a technique employing the rectangular array shown in table 3.1. The successive diameters $D_{1a}, D_{2a}, \ldots, D_{1b}, D_{2b}, \ldots, D_{1c}, D_{2c}, \ldots$ etc are measured for the components A, B, C, . . . etc and their squares written out as shown.

The differences of successive vertical columns give values for Δ_{12}, Δ_{23} etc, while the differences of horizontal rows give values for ${}_1\delta_{ab}, {}_2\delta_{ab}, \ldots, {}_1\delta_{bc}, {}_2\delta_{bc}, \ldots$ etc. With this arrangement gross errors of measurement should be immediately apparent. Some care in interpretation is still required: in any given row the successive values of Δ are not independent and their mean is of course simply derived from the difference of the first and last measured fringe. These in fact are likely to be the most in error—high dispersion near the centre may make the assignment of fringe peak difficult, while the narrow outermost fringe may be subject to distortions of grain, lens aberration and finite number of interfering beams. Various schemes that give due weight to the intermediate fringes have been proposed. Given $2n$ fringes, one of the simplest methods obtains independent estimates of $n\Delta$ from the n differences $D^2_{n+1} - D^2_1$, $D^2_{n+2} - D^2_2, \ldots, D^2_{2n} - D^2_n$.

Table 3.1 Rectangular array for the reduction of fringe diameters.

Component	Ring number 1		2		3		4	
A	D^2_{1a}	Δ_{12}	D^2_{2a}	Δ_{23}	D^2_{3a}	Δ_{34}	D^2_{4a}	. . .
	${}_1\delta_{ab}$		${}_2\delta_{ab}$		${}_3\delta_{ab}$		${}_4\delta_{ab}$	
B	D^2_{1b}	Δ_{12}	D^2_{2b}	Δ_{23}	D^2_{3b}	Δ_{34}	D^2_{4b}	. . .
	${}_1\delta_{bc}$		${}_2\delta_{bc}$		${}_3\delta_{bc}$		${}_4\delta_{bc}$	
C	D^2_{1c}	Δ_{12}	D^2_{2c}	Δ_{23}	D^2_{3c}	Δ_{34}	D^2_{4c}	. . .
	${}_1\delta_{cd}$		${}_2\delta_{cd}$		${}_3\delta_{cd}$		${}_4\delta_{cd}$	
D	D^2_{1d}	Δ_{12}	D^2_{2d}	Δ_{23}	D^2_{3d}	Δ_{34}	D^2_{4d}	. . .
⋮								

A value for Δ may also be obtained from optical parameters. Equation (3.85) may be extended as follows:

$$\Delta = 8f^2/n_a = 8f^2\lambda_a/2d = \sigma_1 8f^2\lambda_a. \tag{3.91}$$

The term $8f^2\lambda_a$ has been called the scale factor C (Jackson and Kuhn 1939); C and Δ may be obtained from accurate measurement of the focal length of the fringe forming lens (plus if necessary the magnification factor k in the auxiliary spectrograph as in figure 3.10); equation (3.91) also permits the evaluation of Δ from narrow fringes recorded at an entirely different wavelength.

The wavenumber shift between components is given in terms of the scale factor C by

$$\Delta\sigma_{ab} = \sigma_1\delta_{ab}/\Delta = \delta_{ab}/C. \tag{3.92}$$

In general, the measurement accuracy of δ_{ab}/Δ rarely exceeds 1 part in 1000; a comparable accuracy for plate separation d_1 and hence σ_1 can often be provided by a good micrometer or set of slip gauges.

In some cases, rather than photographing complete fringes (or sections thereof) it may be more convenient to record one side or the other, possibly not even including the centre. If the centre can be located, analysis can proceed in the usual manner with measured radii in place of diameters. Otherwise, parabolic fitting of fringes is required. For fringes well off-axis various approximate schemes have been discussed (e.g. Tolansky 1947, p133). Consider the fringe positions shown in figure 3.13 for line components A and B. As an approximation the fractional order difference between them was given by McNair (1926) as

$$\delta n = 2\delta_1/(\Delta_{1B} + \Delta_{2A}) = 2\delta_2/(\Delta_{2B} + \Delta_{3A}). \tag{3.93}$$

Tolansky showed that for the 10th fringe off-centre these expressions gave an error in the determination of δn of no more than about 1 in 10^3. If greater

accuracy is required various extensions of the array technique (table 3.1) can be employed.

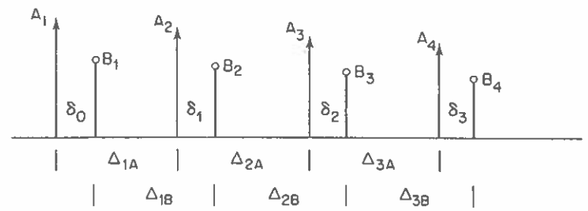

Figure 3.13 Fringe positions well off-axis showing two line components A and B and the measured intervals δ and Δ employed in the approximation of equation (3.93).

(b) *Measurement of wavelength*

Historic measurements of wavelength have been outlined in Chapter 1. A particularly detailed account of earlier work using photographic recording has been given by Meissner (1941). The present section reiterates some of the basic principles. At the centre of the fringe pattern due to a line of wavelength λ_a in air, and for an air-spaced étalon, the usual relationship is

$$(p_a + \varepsilon_a/2\pi)\lambda_a = 2d \tag{3.94}$$

where p_a is the order number of the first fringe and $\varepsilon_a/2\pi$ is the fractional order number. Of these terms $\varepsilon_a/2\pi$ is readily evaluated using the relationships established in the previous section. (For purposes of comparison the refractive index of air is well enough established for different values of humidity, pressure and temperature.) To determine λ_a both d and p_a must be accurately determined; for $d \sim 5$ mm and $\lambda \sim 0.5\ \mu$m, p is 20000. Obviously the exact whole number p must be found, and d established with a precision comparable with the 1 part in 10^3 accuracy or better available in $\varepsilon_a/2\pi$.

An accurate value for d is readily found given two well known wavelengths λ_1 and λ_2 and an approximate value d' for the plate separation, using Benoit's method of exact fractions. From equation (3.94) we have

$$(p_1 + \varepsilon_1/2\pi)\lambda_1 = (p_2 + \varepsilon_2/2\pi)\lambda_2 = 2d \tag{3.95}$$

where p_1 and p_2 are the exact (integral) order numbers of the first bright rings. From the approximate value d' for the plate separation a first estimate p'_1 may be calculated. A range of integral values $p'_1 \pm i$ may then be inserted in equation (3.95) to give a corresponding set of values $p'_2 \pm k + \varepsilon_k/2\pi$ (k of course being an integer). In general the ε_k will differ from ε_2, but one of them should be within measurement error such that

$$(p'_1 + n + \varepsilon_1/2\pi)\lambda_1 = (p'_2 + m + \varepsilon_2/2\pi)\lambda_2. \tag{3.96}$$

If these values are sufficiently close to $2d'$, within the limit of error of the original estimate for the separation, one may safely assume that

$$p_1 = p'_1 + n \quad p_2 = p'_2 + m \quad \text{and } d = (p_1 + \varepsilon_1/2\pi)\lambda_1/2. \tag{3.97}$$

In general a check with other known standard lines would be advisable; for well separated lines correction for different phase changes at the interferometer surfaces is required, and also for dispersion in air (see e.g. Meissner 1941).

Returning to equation (3.94) the wavelength of an unknown line is readily determined provided that its wavelength λ_x is known within 0.1 Å (from e.g. grating measurements). With d known exactly from the measurement with two standards, the ratio d/λ_x may thus be estimated with sufficient precision to establish the correct integral value of p_x (at least for relatively small étalon spacers and visible light). Any remaining uncertainty would be removed by measurement with a second, known, spacer.

It cannot be denied that accurate wavelength measurement with photographic methods proved difficult and time consuming. Harrison in 1946 wrote:

> 'The paucity of wave-length standards has resulted principally from the difficulty of making measurements with the Fabry–Perot interferometer, not so much because of the delicacy of adjustment and control required for its proper use, as because of the tedious and involved computations needed to determine wave-length values once the interferometer patterns have been measured.'

Harrison went on to describe a new type of measuring and computing engine to determine wavelengths and wavenumbers directly from spectrograms made with an interferometer and concave diffraction grating. The interference pattern (photographic negative) of each line was projected on a screen carrying a set of parabolic fiducial lines drawn to fit the optical system. The operator then moved the screen until the lines coincided with the density maxima of the pattern, as shown in figure 3.14. The fractional order of interference $\varepsilon/2\pi$ at the centre could then be read to three figures from a dial. Corrections for dispersion of air and phase change variations with wavelength were introduced with mechanical cams. Wavelength and wavenumber could be read off dials correct to seven or eight figures depending on the line sharpness, giving a precision of a few ten-thousandths of an angstrom and a claimed increase in speed over previous methods of at least 20-fold.

A good example of a modern computer controlled wavemeter, of design following Byer *et al* (1977), has recently been described by Fischer *et al* (1981) and is shown in figure 3.15. Detection of fringe patterns is by linear silicon diode arrays with 1024 diodes. Analysis proceeds in an identical manner as that described for photographic use. Computation of ring diameters with least-squares fitting provides values of the fractional order δn or $\varepsilon/2\pi$ with uncertainty < 0.01. The integral order number for the smallest étalon (free spectral range 1000 GHz) could be accurately determined by the first wavelength estimate provided by the small grating. This then gave a second wavelength estimate for determining the exact integral order number in the middle étalon,

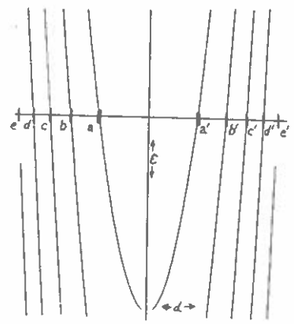

Figure 3.14 Interference fringes shown projected on a screen carrying parabolic fiducial lines. In this example (from Harrison 1946) the operator has made a setting for fringes a, b, c′ and d′. Because of an unsatisfactory projecting lens (later replaced) other fringes are slightly misplaced. Reproduced by permission of the Optical Society of America.

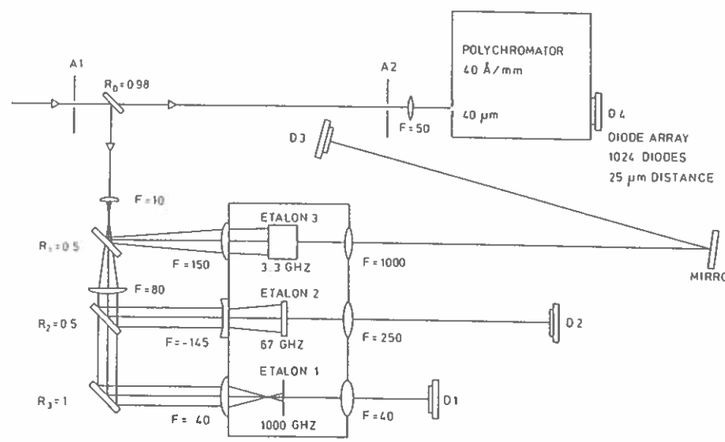

Figure 3.15 The optical layout of the wavemeter from Fischer *et al* (1981): D1–D4 are linear silicon diode arrays. Beam divergence is selected by the lenses F; 5–6 fringes are illuminated on the arrays D1–D3. The solid, fuzed silica étalons are of thickness 0.1, 1.5 and 30 mm. Reproduced by permission of North-Holland Physics Publishing.

and so on to the largest étalon (free spectral range 3.3 GHz). The étalons were in fact of solid fused silica; careful calibration with standard lines was required and knowledge of the wavelength-dependent refractive index $\mu(\lambda)$. The ultimate accuracy of measurement was determined by the thickest étalon and amounted

to better than 70 MHz (0.002 cm^{-1}). This could have been considerably improved with a thicker, vacuum-spaced étalon. For pulsed lasers a single pulse with a minimum energy of 5×10^{-6} J was sufficient for measurement. The computing time for absolute wavelength was about 20 s; line profiles and general wavelength stability, mode structure etc could be monitored instantaneously on a screen display. This is well illustrated by the fringes shown in figure 3.16.

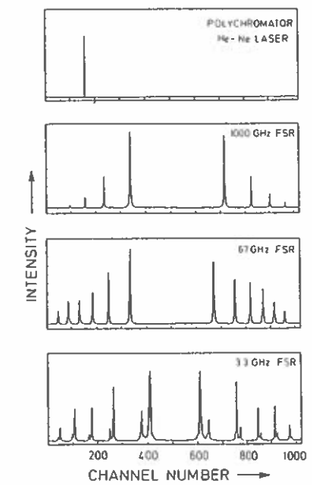

Figure 3.16 Signals recorded by the diode arrays D1–D4 of figure 3.15. The illumination was a He–Ne laser at 632 nm. The two fundamental modes, spaced 500 MHz apart, are resolved only in the lowest trace due to the thickest étalon. (From Fischer *et al* 1981.) Reproduced by permission of North-Holland Physics Publishing.

3.6 Plate defects

3.6.1 General considerations

To attain high resolving power it is obviously necessary that the interferometer surfaces should be of the highest possible quality—smooth, flat and set accurately parallel to one another. The recognition that surface defects could provide a limit on attainable resolution became evident in the late 1920s with the improvement in reflectivity of metallic, particularly silver, films. Nowadays optical plates individually flat to $\lambda/150$ are readily obtained; pairs of plates may be purchased, at a premium, matched to better than $\lambda/200$ over 5 cm or more diameter. The appropriate reflective coating of such high-quality plates demands careful consideration; additionally, of course, multilayer coatings (§2.6) must be

at least of the same uniformity if the full capabilities of the plates are to be realized.

The first attempt to investigate the problem quantitatively appears to have been made by Burger and Cittert in 1927; their analysis was considerably extended by Dufour and Picca (1945) and has since been followed by several other authors including Chabbal (1953), Hill (1963), Hernandez (1966, 1970b), Roesler (1969), Ramsey (1969), Dunaev *et al* (1977), Prikryl (1984) and Sloggett (1984). In nearly all these treatments the non-ideal interferometer, incorporating departures from flatness and parallelism, is treated as being made up of sets of primary interferometers all of different plate separation. The resulting intensities due to each of these sets are then summed incoherently to produce a final instrumental function. However this procedure has been criticized by Steel (1967) on the grounds that, rigorously, the amplitude contribution from each primary interferometer should be summed in a coherent manner and the intensity distribution derived from this coherent summation. In practice the differences are rather small and it is in fact difficult to point to more than a handful of spectroscopic experiments where corrections for plate defects have either been applied (e.g. Vaughan 1966) or incorporated (e.g. Killeen and Hays 1984).

Either way one first requires to consider models of possible plate defects and from these derive the area distribution $A(d)$ of plate separation. Figure 3.17 shows some possibilities in which the defects have been projected into one surface and the other may be considered as ideally flat and smooth. Following Dufour and Picca and other authors we suppose the mean spacing between the plates to be d with a maximum excursion $\pm\,\Delta_M$. The surface area of the set of interferometers with spacing between $d + \Delta$ and $d + \Delta + \delta\Delta$ is then supposed given by $\delta S = D(\Delta)\delta\Delta$. The phase decoupling for interference at the spacing $d + \Delta$ compared with the mean spacing d is given by

$$\delta\varphi = 4\pi\Delta/\lambda. \tag{3.98}$$

Following incoherent addition of intensities the resultant instrumental form may be simply shown to be the convolution of the ideal Airy function and the surface distribution function $D(\delta\varphi)$ expressed in terms of phase decoupling $\delta\varphi$.

3.6.2 Different types of defect

The various forms of $D(\delta\varphi)$ shown in figure 3.17 are worth discussing individually.

(i) Figure 3.17(a) shows a dish-like aberration with spherical curvature. For circular plates the area δS between Δ and $\Delta + \delta\Delta$ is given by $2\pi r\delta r$ where r is the radius to Δ as shown. If the radius of curvature of the dish-like plate is e, differentiating the relationship $(2e \times \Delta) = r^2$ gives $|\delta\Delta| = 2r\delta r$. Consequently $D(\varphi)$, equal to $\delta S/\delta\Delta$, is of constant value within the range of phase $\pm\delta\varphi_M$, as shown by the 'top hat' distribution in the figure. Approximations to this type of defect are not uncommon—plates are usually of somewhat poorer figure towards the edges compared with the centre. In addition faulty mounting of the plates

and resultant strain can induce curvature in one if not two dimensions. Obvious defensive measures are to reduce the available aperture in the one case, and to check and recheck the assembly of the interferometer in the other.

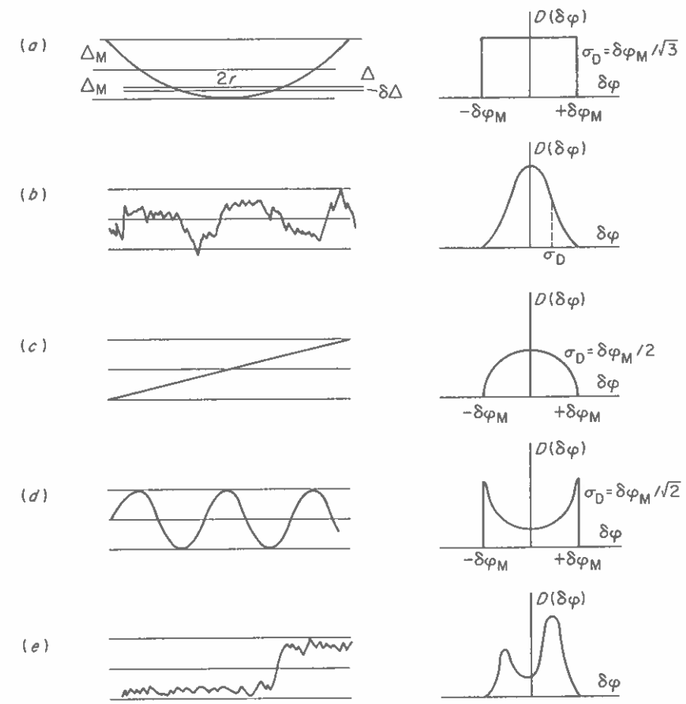

Figure 3.17 A schematic representation of various plate defects in which the plate separation has been projected into one surface (the other being considered ideally flat and smooth). The corresponding distributions of phase variation $D(\delta\varphi)$ for circular plates are also shown (see discussion in text) and, where appropriate, the standard deviation σ_D of the distribution.

(ii) Figure 3.17(b) shows flat plates with random plate defects of small scale. If these are supposed to be randomly distributed the surface distribution may readily be shown to be of Gaussian form (Chabbal 1953). In this case if the standard deviation σ_D over the distribution is equal to λ/q, (where obviously the larger q is the better the plate quality) the defect distribution is a Gaussian function of width (FWHH) equal to $1/N_D$ of an order, where $N_D = q/4.7$. Clearly

N_D represents a limiting 'plate defect' finesse. This type of defect is probably not so troublesome nowadays. The demand for very low scatter-loss surfaces for laser usage has led to great improvements in micro finish, stemming for example from prolonged polishing.

(iii) Figure 3.17(*c*) represents a lack of parallelism of otherwise ideally flat plates. The form of $D(\Delta)$ may be simply shown to be hemispherical. If the difference of plate separation from one side to the other is equal to λ/p the width (FWHH) of the profile is $(p\sqrt{3})/4$.

(iv) Figure 3.17(*d*) shows an undulating character on otherwise smooth plates. This type of defect has been discussed in particular by Hill (1963) and Dunaev *et al* (1977) in one and two directions on the surface of the plate. In general the surface distribution function will have the form shown with peaks at the extremes of phase difference. This type of defect is now likely to be more apparent with improvements in surface micro finish. It is potentially rather dangerous in that for plates of very high reflectivity finesse the resultant instrumental contour may become asymmetric or even exhibit subsidiary maxima.

(v) Figure 3.17(*e*) shows in rather exaggerated fashion a step defect. As such it would be readily recognized. However a small defective region perhaps towards the edge of a pair of plates might not be found except by careful checking. It can lead of course to the same potential problems as the last category: asymmetry and satellites.

Where appropriate the standard deviation σ_D for the different distributions is shown in figure 3.17. Sloggett (1984) has suggested that the half-value width W (FWHH) of a defect broadened fringe may be given approximately as $(W_0^2 + \alpha^2\sigma_D^2)$. From computed convolutions (see the next section) of different defects with the Airy distribution, treated as a Lorentzian, he offers $\alpha^2 \simeq 12$ as a reasonably good approximation for most defects and in some cases for quite large values of σ_D/W_0, where W_0 is the width of the undistorted fringe.

In the analysis of Steel (1967, p119) the convolution of D was made with the amplitude expression (equation (3.3)) and for small departures from flatness a power series approximation was used. Steel noted that this approach was used to derive optical tolerances for image forming systems. From the mean value of the amplitude across the aperture the transmittance of the interferometer is obtained in terms of the mean square phase variation $\overline{\delta\varphi^2}$ across the plates. Steel gave the result that the modified instrumental finesse N_I was related to the ideal reflectivity finesse N_R by

$$N_I \simeq N_R\,(1 - 0.2\,\overline{\delta\varphi^2}N_R^2) \tag{3.99}$$

and the peak transmittance was reduced by the factor

$$1 - \overline{\delta\varphi^2}[(1 + R)^{1/2}N_R/\pi]^2. \tag{3.100}$$

Steel compared this with the usual expression for a simple image forming system and showed that, for an interferometer considered as a component of such a system, the usual optical tolerances should be divided by

$$(1 + R)^{1/2}N_R/\pi \simeq 0.4N_R. \tag{3.101}$$

In a sense all these discussions are somewhat academic in that the manufacturers specification usually states that the plates show defects less than λ/q over a specified diameter, and rarely gives a functional form to the distribution of defects. However it may be noted that advances in technique now permit this (e.g. Meaburn *et al* 1976). Also Killeen *et al* (1981) have constructed parallelism contours with an array of 48×48 individual measurements over a 35 mm diameter étalon—see §4.4.2 and Appendix 11.

Nevertheless the question is usually what reflectivity finesse should be sought and what corresponding value of reflectivity should be specified. The answer must depend not only on the plate quality but also on the experimental problem in view: the likely brightness of the source, whether the measurement is a single-line profile or of a many-component spectrum, the expected asymmetry, etc. Obviously a choice of $N_R \sim q$ leads to an instrumental profile strongly influenced if not dominated by plate defects (especially for cases (iv) and (v) above, as is also evident from equation (3.99)). A general concensus would suggest that a suitable value of N_R lies in the range $q/2$ to $q/4$. Thus with plates flat as a pair to $\lambda/200$ a reflectivity finesse of 70 or more might be sought; for plates flat to $\lambda/100$ it might not seem worth aiming above $N_R \sim 50$. The resultant instrumental function is still of course determined by the combination of plate reflectivity and defect. For a well defined problem, e.g. the measurement of a single Doppler broadened and shifted line, a very precise analysis of optimum plate characteristics and most particularly the required reflectivity finesse may be made, as shown in the valuable and comprehensive discussion of just this problem by Hernandez (1979) which is outlined in Appendix 10.

As a broad summary of this section one may conclude that while the experimentalist needs a good appreciation of the surface quality of an interferometer, in practice this is just one amongst many factors that determine the actual instrumental profile. Particularly with the advent of dedicated microprocessors and large-scale data handling, accurately measured instrumental profiles can be rapidly incorporated into the analysis of experimental spectra. An early example of this was given by Best (1967). More recently Killeen and Hays (1984) have described a quasi-linear least-squares fitting procedure involving matrices for rapid reduction of data for the multichannel interferometer on the Dynamics Explorer spacecraft (see also Appendix 10 and §7.3).

3.7 Profile analysis of experimental spectra

3.7.1 Folding and deconvolution

In constructing an experimental spectrum $E(\nu)$, the energy from the source within a narrow-frequency band $\delta\nu'$, and given by $S(\nu')\delta\nu'$, is redistributed by the instrumental function $I(\nu - \nu')$. Thus $E(\nu)$ is given by the well known folding or convolution integral

$$E(\nu) = \int_{-\infty}^{+\infty} S(\nu')I(\nu - \nu')\delta\nu'. \tag{3.102}$$

Ordinarily one may assume that the instrumental function does not change with ν' so that the process of transformation (from ideal source spectrum to actual experimental spectrum), defined by the above equation, is both linear and invariant. This class of folding integral has several important properties; firstly it may be shown to be symmetric by simple change of variable, so that

$$\int_{-\infty}^{+\infty} S(\nu')I(\nu - \nu')\delta\nu' = \int_{-\infty}^{+\infty} S(\nu - \nu')I(\nu)\delta\nu'. \tag{3.103}$$

Secondly the Fourier transform of the convolution of two such functions is equal to the product of their individual Fourier transforms. In an obvious notation this may be represented as

$$\tilde{E}(k) = \tilde{S}(k) \times \tilde{I}(k) \tag{3.104}$$

where the transformation is made to the reciprocal 'k-space'. The folding integral is often written as $E = S*I$. From the Fourier transform relationship above, if the instrumental function is itself made up of the convolution of several functions $X*Y*Z$, then

$$E = S*X*Y*Z. \tag{3.105}$$

The experimental problem is thus simply stated: from the observed experimental spectrum $E(\nu)$ and given the instrumental profile $I(\nu)$ (which at least in principle should be obtainable with very high precision) how does one best obtain the ideal source spectrum $S(\nu)$? This problem has been discussed at great length in many areas of physics. At first sight it would seem from equations (3.103) and (3.104) rather straightforward to form the Fourier transforms $\tilde{E}(k)$ and $\tilde{I}(k)$, divide them to give $\tilde{S}(k)$ and perform an inverse transform of the quotient to derive $S(\nu)$. For well behaved analytic functions, mathematical deconvolution works without difficulty; for real spectra with random noise the relative error rapidly becomes larger for higher Fourier terms—larger values of k—and the noise eventually dominates. This is clearly illustrated in figure 3.18; the transform of instrumental functions $\tilde{I}(k)$ often in practice falls actually to zero (e.g. the triangular transform due to the sinc^2 instrumental function of the diffraction grating) and always approaches zero at higher k-values of the order of $(\Delta\nu_{1/2})^{-1}$ (note the negative exponential transform of the Lorentzian function and the Gaussian Fourier transform of the Gaussian function). In consequence, on multiplying the instrumental transform $\tilde{I}(k)$ with that of the intrinsic source spectrum $\tilde{S}(k)$, information is irretrievably lost in the resultant experimental transform $\tilde{E}(k)$ for zero values of $I(k)$. For small non-zero values the procedure is very dependent on the noise characteristics. This is particularly evident on dividing $\tilde{E}(k)$ by $\tilde{I}(k)$ to form $\tilde{S}(k)$; at larger k experimental data points will be divided by very small numbers. More or less precise deconvolution is still possible in principle for high-quality spectra; however the procedure is readily vitiated by noise. Again at first sight the solution would seem to be to exclude the larger k-values. This however is of course equivalent to supposing that the

only interesting information lies in the low-frequency structure; the fine structure relating to the higher frequencies is excluded, and investigation of this is likely to have been the principle aim of the experiment in the first place!

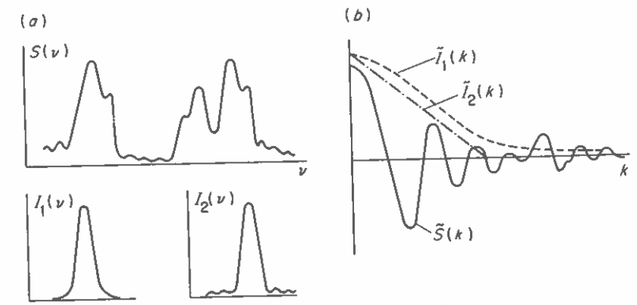

Figure 3.18 The deconvolution problem. (*a*) The intrinsic source spectrum $S(\nu)$ and instrumental functions $I_1(\nu)$ and $I_2(\nu)$. (*b*) Corresponding Fourier transforms. The observed experimental transform $\tilde{E}(k)$ is given by the product of $\tilde{S}$ and $\tilde{I}$: with $\tilde{I}_1(k)$ information is rapidly attenuated for higher k-values and is only recoverable for spectra of very high signal-to-noise; $\tilde{I}_2(k)$ falls to zero and high-frequency information is lost in the experimental transform.

Faced with this dilemma several methods of deconvolution have been introduced over the years, some of which have apparently rather little to do with Fourier transformation. These include the iterative procedure of Burger and van Cittert (1932), the Stokes method after a Fourier series approximation to $E(\nu)$, graphical methods (Lloyd 1969) and a linear equation method (Louer *et al* 1969) (see also Hill and Ioup 1976). Analysis shows however that they are mathematically all equivalent to Fourier transformation in which the required cut-off at higher k-values is applied in a more or less smooth or abrupt manner. Their relative attractiveness will depend on available computing facilities and the general inclination of the experimentalist. Several papers have described individual methods and include Cooper (1971), Hays and Roble (1971) with a technique of Fourier decomposition for recovering single Doppler profiles, and Zipoy (1979) with a Fourier based inversion algorithm for multi-component spectra. Faced with the problem of analyzing a complex experimental spectrum of unknown form and content one can draw at least two guidelines: (i) any technique of deconvolution should be questioned as to what criteria it offers for distinguishing between noise and genuine fine structure, and (ii) any components or structure that appear in the deconvoluted spectrum that are not clearly evident in the experimental spectrum should be regarded with extreme suspicion. These reservations would perhaps suggest that, rather than attempt

deconvolution, the experimentalist would in fact do better to compare experimental spectra with theoretical spectra convolved with the instrumental function.

Fortunately this rather perplexing situation is fully manifested only in those rather rare occasions when little or no '*a priori*' information exists as to the nature of the spectrum under study. The fact is that usually the general form and content (shape, number of components etc) will be known and the problem becomes one of measurement of width, shift and assignment of experimental error for a well recognized spectrum. In this case a common analytical procedure is to compare the observed spectrum with a series of constructed spectra in which component width and separation are adjustable parameters (e.g. Hühnermann 1967, Larson and Andrew 1967, Day 1966, Hildum 1971, Goorvitch 1977). Various criteria e.g. minimization of variance between the experimental and theoretical curve (with suitable weighting of 'important' parts of the profile?) may be employed for selection of a 'best' fit. Remarkable precision for fitting to a single experimental recording has often been claimed. This too needs to be regarded with considerable scepticism—effects of e.g. slight non-linearity or non-reproducibility of frequency scale, source brightness fluctuations etc may introduce experimental errors one or two orders of magnitude greater than those evaluated from a 'best fit' to a single recording. An obvious precaution is a careful comparison of the analyses of many recordings and, if necessary, suitable improvement of the spectroscopic technique to improve consistency. This is apart, of course, from the proper evaluation of possible systematic errors other than purely 'fitting' errors.

3.7.2 Single profiles—parameter estimation

The situation is rather different again in the analysis of a single, well separated, spectral profile. In many cases the physical contributions to the breadth of the line will be well known e.g. Doppler effect, pressure broadening, phonon attenuation, spread of scattering vector etc at the source, plate reflectivity and defects, and scanning aperture at the interferometer. Correspondingly, the analytic form due to many of these mechanisms—Gaussian, Lorentzian, top hat, Airy functions etc—will also be well known. Analysis of the experimental line profile may then proceed on the assumption that it is made up of an appropriate set of such functions and with the aim of determining the parameters, usually the half-value width of each component. This procedure is simplified if the contributing functions can be reduced to or approximated by just two types: Gaussian and Airy, or Lorentzian and Gaussian for example. Denoting these two 'known' functions as Y and Z of separate half-value width y, z and their convolution X of half-value width x, these may be represented as

$$X = Y * Z \quad \text{and } x = y \oplus z. \tag{3.106}$$

If $\langle x \rangle$ and $\langle z \rangle$ are values measured from experimental and instrumental profiles, then an estimate $\langle y \rangle$ may be immediately determined from graphs or look-up tables derived from trial convolutions of Y and Z. This procedure is illustrated in Appendix 9 for Voigt functions (convolutions of Lorentzian and Gaussian functions). These types of technique may also be extended to the

situation where neither y nor z is known or measurable separately. If however their functional form is well established, their relative magnitude is determined uniquely by a single *shape* measurement of the convolved spectrum. This measurement might be a ratio of profile widths at two fractions of the peak intensity, or a ratio of profile heights at different width. These methods are sometimes referred to as 'deconvolution' but should more correctly be described as 'parameter estimation' from width and shape measurements. Such simple techniques are often directly applicable to recorded spectra and can provide an accuracy of 5 to 10%. This may give a useful starting point for more detailed computer fitting with iterative or matrix procedures. In many cases however a precision of 5–10% may be adequate; in practice it is often remarkably hard to improve on, and a realistic experimental appraisal may indeed show that such an endeavour would not be justified.

In practice Voigt functions are often required (Appendix 9); many evaluations of Voigt functions and their application to the analysis of spectral line profiles have now been made (e.g. van de Hulst and Reesink 1947, Elste 1953, Posener 1959, Godoli and Santoro 1960, Davies and Vaughan 1963, Tako and Ohi 1965, de Vries 1965, Larson and Andrew 1967, Leidecker and La Macchia 1967, Reichel 1968, Whiting 1968, Jansson 1970, Jones and Misell 1970, Ivanov and Fishman 1973, Ekstrom 1973, Kielkopf 1973, 1979, Wertheim *et al* 1974, Drayson 1976, Schmidt 1976, Pierluissi and Vanderwood 1977, Hui *et al* 1978, Puerta and Martin 1981, Guilpin and Mangeot 1982). These papers deal in varying degrees of sophistication with the precise form of the instrumental function. Another early detailed analysis was due to Bayer-Helms (1963), while Ballik (1966) provided a table of profiles giving the intensities of Fabry–Perot fringes at different fractions of the free spectral range. Drayson (1976) and Pierluissi and Vanderwood (1977) gave specific FORTRAN IV computer programs for the evaluation of Voigt functions (see also Hui *et al* 1978). Kielkopf (1979) considered the problem of deconvolution for asymmetric spectral lines. By using the high extinction of a triple-pass interferometer (Chapter 6) Bradberry and Vaughan (1977a) were able to measure central Lorentzian components 100 times narrower than the instrumental function. More recently Vanderwal *et al* (1981) employed iterative methods to demonstrate the measurement of Brillouin components as small as one-sixth of the instrumental width. As already noted the methods of Hays and Roble 1971 have been considerably developed for measurements in the upper atmosphere (see e.g. Appendix 10 and the paper of Killeen and Hays (1984) which are further discussed in §7.3).

The reader may wish to consult the above papers in evaluating analytical techniques appropriate to his or her needs. Any technique should be used with caution and a realistic appraisal of available precision. In conclusion it is worth noting several experimental defects that could lead to varying degrees of profile distortion and unexpected loss of accuracy: (i) spatial and temporal intensity modulation across the source; (ii) uneven illumination of the interferometer; (iii) poor focusing of the collimating lens; (iv) maladjustment of the interferometer; (v) imprecise focusing of the fringe forming lens; and (vi) inaccurate centring of the scanning aperture on the circular fringes. Of these, (v) and (vi) are particularly severe and lead rapidly to spurious broadening and asymmetry.

The asymmetric instrumental function for such an off-axis scanning aperture is shown in figure 3.19. This case has been worked out in detail by Hernandez (1974) who also evaluated the related problems of (i) scanning of photographic fringes by densitometry, (ii) vidicon scanning of fringe patterns and (iii) spectral scanning by tilting interference filters. Accurate centring and focusing may be carried out in practice (a) by precise visual observation on a screen, (b) by monitoring fringe profiles on an oscilloscope or (c) with a series of photographic exposures (see the discussion in §4.4.3). It is also worth reiterating that experimental instrumental functions should be recorded under exactly the same conditions of illumination as arranged for experimental spectra. For example merely passing a narrow laser beam into the interferometer may produce a line profile considerably different from the true instrumental function appropriate to actual conditions of spectroscopic use.

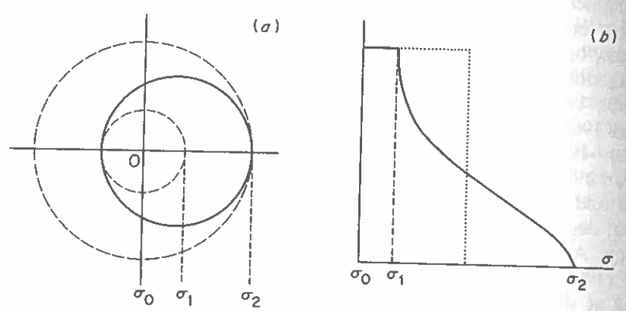

Figure 3.19 The aperture instrumental function for an inaccurately centred pinhole aperture. (*a*) The aperture is shown by a full curve and the centre of the fringe system is at 0. The wavenumbers σ_0, σ_1 and σ_2 at different radii (relative to 0) are shown. (*b*) The corresponding aperture instrumental function (full curve). Note the gross asymmetry—see the text and detailed evaluation of Hernandez (1974). The dotted line shows the rectangular instrumental function for the aperture centred correctly.

3.8 Summary of useful expressions

(i) The geometric phase lag φ between successively reflected beams is

$$\varphi = 2\pi(2\mu d\cos\theta)/\lambda_0. \tag{3.1}$$

(ii) The Airy intensity transmission coefficient is

$$I_T = T^2/(1 + R^2 - 2R\cos\varphi) = [T^2/(1 - R)^2]\mathcal{A}(\varphi) \tag{3.8}$$
$$= [T/(1 - R)]^2/[1 + F\sin^2(\varphi/2)]$$

where $F = 4R/(1 - R)^2$, φ is taken to include any surface phase changes and

$\mathcal{A}(\varphi)$ is the Airy shape function equal to $(1 + F\sin^2(\varphi/2))^{-1}$.

(iii) For identical high-reflective coatings the reflected transmission coefficient is

$$I_R = (R + T) - [\mathcal{A}(\varphi)/(1 - R)^2]T[1 - R(R + T)]. \tag{3.26}$$

(iv) In transmission the angular radius of the nth-order fringe is given by

$$\theta_n = \cos^{-1}(n\lambda_0/2\mu d') \tag{3.34}$$

where $\mu d'$ is the optical path separation of the plates and includes any phase changes on reflection. The radius of the nth-order fringe is given by $f\theta_n$ and the differences of squares of the radii of successive fringes is $2f^2/n$.

(v) The half-width of a fringe at half-intensity is given by $2/F^{1/2}$ rad. In equation (3.8) above the reciprocal of the full width at half-intensity expressed as a fraction of an order is called the reflectivity finesse N_R and is given by

$$N_R = \pi\sqrt{R}/(1 - R) = (\pi/2)\sqrt{F}. \tag{3.37), (3.52}$$

(vi) The transmission at the peak of a fringe is given by

$$I_{max} = I_0[T/(1 - R)]^2 = I_0[1 - A/(A + T)]^2 \tag{3.38}$$

and at the minimum is

$$I_{min} = I_0[T/(1 - R)]^2/(1 + F)$$
$$= I_0[T/(1 + R)]^2. \tag{3.40b}$$

(vii) The extinction or contrast ratio I_{max}/I_{min} is equal to

$$(1 + R)^2/(1 - R)^2.$$

(viii) An alternative series form of the intensity transmission function is

$$I_t = [T^2/(1 - R^2)]\left(1 + 2\sum_{m=1}^{\infty} R^m\cos m\varphi\right) \tag{3.46a}$$

$$= [T^2/(1 - R^2)]\sum_{m=-\infty}^{m=+\infty} R^{|m|}e^{im\varphi}. \tag{3.46b}$$

(ix) The free spectral range is the frequency or wavenumber interval between two spectral lines which would overlap exactly in successive orders and is given by

$$\sigma_1 = 1/2\mu d \quad \text{cm}^{-1} \tag{3.55}$$

or

$$\nu_1 = c/2\mu d \quad \text{Hz}.$$

(x) The resolving limit is given by

$$\Delta\sigma_{min} = \sigma_1/N_R \quad \text{cm}^{-1} \tag{3.59}$$

and the limiting resolving power $\mathcal{R}_0$ by

$$\mathcal{R}_0 = nN_R. \tag{3.62}$$

(xi) The diameter of the 'first full fringe' is given by

$$a_{2\pi} = 2f\theta_{(1)} = 2f\sqrt{\lambda/d} \tag{3.77}$$

and a physical aperture of diameter a at the centre of the fringe pattern will transmit the fraction $(a/a_{2\pi})^2$ of the free spectral range.

(xii) In photographic recording the difference of fractional order ε_c between two spectral lines a and b is given by

$$({}_a\varepsilon_c - {}_b\varepsilon_c) = \delta_{ab}/\Delta \tag{3.87)–(3.90}$$

where δ and Δ are given by the successive ring diameters D:

$$\delta_{ab} = {}_aD_p^2 - {}_bD_p^2 \quad \text{and } \Delta = {}_aD_{p+1}^2 - {}_aD_p^2 = {}_bD_{p+1}^2 - {}_bD_p^2.$$

(xiii) The experimental spectrum $E(\nu)$ is given by the fold or convolution integral of the source spectrum $S(\nu)$ and instrumental function $I(\nu)$:

$$E(\nu) = \int_{-\infty}^{+\infty} S(\nu')I(\nu - \nu')\delta\nu'. \tag{3.102}$$

The comparable relationships between their individual Fourier transforms is

$$\tilde{E}(k) = \tilde{S}(k) \times \tilde{T}(k). \tag{3.104}$$

4 Practical Operation of Plane Fabry–Perot Interferometers

4.1 Introduction

In this chapter, technical aspects of the design and operation of plane Fabry–Perot interferometers in the practical modes of operation are discussed. Table 4.1 outlines the range of various experimental parameters over which Fabry–Perot interferometric spectroscopy has been applied in the visible and near-infrared region. (Techniques and application in the far-infrared have recently been authoritatively reviewed by Chantry (1982). Specific discussion of this regime, involving as it does rather different considerations, is reserved for Chapter 10.)

Table 4.1 Range of experimental parameters.

(*a*)	Required resolution: $\gtrsim 1$ Å ($\sim 10^{-2}$ GHz): plate separation of few tens of micrometres	—— × $\sim 10^5$ ⟶	$\lesssim 1$ MHz; plate separation several tens of centimetres
(*b*)	Experimental period: $\lesssim 10^{-6}$ s; transient phenomena, e.g. plasmas etc	—— × 10^{10} ⟶	$\gtrsim 10$ h; low-intensity sources, e.g. light scattering spectroscopy
(*c*)	Operating constraints: Fixed free spectral range and plate separation for full working life, e.g. spacecraft operation	—//—⟶	Change of free spectral range and plate separation required several times per day, e.g. light scattering spectroscopy
(*d*)	Nature of spectrum: Single-line, single component of well known form	—//—⟶	Multi-line, many components of unknown form and widely differing intensity